Hierarchical IPTC Topic Classification of French Press Articles with EuroBERT and Hierarchy-Guided Contrastive Learning

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Acknowledgements

Abstract

Abstrakt monitors the press for its clients, and each article needs the topics of the IPTC Media Topics tree, 3 levels deep. An LLM can tag an article, but not at the scale of the service, so this internship built a small model that does it on 1 GPU. The model is a EuroBERT encoder with a head that gives 1 score per topic. I rebuilt HGCLR, a method that teaches the encoder the tree through a contrastive loss, on EuroBERT, and added 4 changes: attention pooling, an article-to-topic attention block, a deeper head, and the asymmetric loss. I designed the avg-max rule as well, a variant of the C-HMCNN max rule that keeps the scores coherent along the tree and trains inside the loss. No real article may serve as training data, so LLMs wrote the training articles and voted on their topics; the test set is real French press. I explored the design on a small data set, then trained the final model on the team's gold set. An ensemble of 3 seeds is the final model. The error analysis shows that the weakest topics are the rarest ones, that a large part of the measured recall gap is noise in the LLM topics, and that real press carries another mix of topics than the generated articles. The report closes with the limits of the work and what I learned.

Contents

Acknowledgements i

Abstract ii

1 Introduction 1
- 1.1 Abstrakt and AI-assisted press monitoring . . . 1
- 1.2 The mission: a topic tag on each article . . . 1
- 1.3 Objectives and constraints . . . 2
- 1.4 Contributions of the internship . . . 2
- 1.5 Outline of the report . . . 3

2 Background 4
- 2.1 The IPTC Media Topics tree . . . 4
- 2.2 Hierarchical multi-label classification . . . 4
- 2.3 Encoders for French press . . . 5
- 2.4 Coherent predictions: C-HMCNN and the max rule . . . 5
- 2.5 Losses for imbalanced multi-label data . . . 5
- 2.6 Evaluation . . . 6

3 Data: 3 data sets 7
- 3.1 Starting point . . . 7
- 3.2 The easy data set . . . 7
- 3.3 The gold training set . . . 8
- 3.4 The real test set . . . 8
- 3.5 Purpose and limits . . . 8

4 Method 9
- 4.1 Model overview . . . 9
- 4.2 Baselines . . . 9
  - 4.2.1 Frozen encoder and an MLP head: choosing the encoder . . . 11
  - 4.2.2 Training the encoder: warm-up and learning rates . . . 12
  - 4.2.3 Graphormer label attention . . . 12
  - 4.2.4 Several seeds: the spread and the ensemble . . . 13
- 4.3 The model block by block . . . 13
  - 4.3.1 Topic features and the graph encoder . . . 13
  - 4.3.2 Positive-sample generator and the 2 views . . . 14
  - 4.3.3 Article-to-topic attention and token FFN . . . 15
  - 4.3.4 Attention pooling . . . 15
  - 4.3.5 Head . . . 15
  - 4.3.6 Classification loss . . . 16
  - 4.3.7 Contrastive loss and the total loss . . . 16
- 4.4 Coherence rules . . . 16

4.4.1 Rules at inference . . . 17
4.4.2 Rules in training . . . 17
4.4.3 The avg-max rule and its implementation . . . 18
4.5 Training settings . . . 18

5 Experimental setup **19**

6 Results **20**
6.1 Baselines on the easy data set: encoders, frozen and fine-tuned . . . 20
6.2 Losses and coherence rules on the easy data set . . . 20
6.3 HGCLR and our changes on the easy data set . . . 21
6.4 Summary on the easy data set . . . 22
6.5 Runs on the gold training set . . . 22
6.6 Choice of the coherence rule . . . 23
6.7 Seeds and the ensemble . . . 24

7 Error analysis **25**
7.1 The errors by level . . . 25
7.2 Errors on the validation data: the model or the LLM topics? . . . 25
7.3 Errors on the real test set: a different mix of topics . . . 26

8 Limits **29**

9 Side work: relevance rerankers for thematic search **30**
9.1 Problem and data . . . 30
9.2 zerank cross-encoders run locally . . . 30
9.3 GLiNER2 as a zero-shot classifier . . . 30
9.4 Findings . . . 30

10 Conclusion **31**

A Method details **34**
A.1 The Graphormer label-attention baseline . . . 34
A.2 Cheaper ways to train the encoder: top layers and LoRA . . . 35
A.3 The avg-max rule: gradient and implementation . . . 35
A.4 Training settings . . . 35

B Metric definitions **37**

C Full architecture graph **39**

List of Figures

2.1 The idea of ASL, 2 figures from Ridnik et al. [11]. Top: the loss on a negative against its score, for cross-entropy, focal loss [9], and focal loss with the margin m. Bottom: the mean confidence in the right answer, p_t, of the positives and of the negatives during training on MS-COCO, with cross-entropy, focal loss and ASL. 6

4.1 The model: the data flow from the inputs on the top row to the topic tags. White: inputs. Blue: HGCLR blocks. Green: our changes. Green with an orange border: avg-max, our variant of the C-HMCNN rule. Solid: the inference path, one pass per article. Dashed: training only. The letters on the arrows are the symbols of the equations in Section 4.3; the positive view $\hat{e}$ follows the same chain as the article and gives $\hat{z}$ and $\hat{s}$. The losses are in Section 4.3; the full graph with tensor shapes is in Appendix C. 10

4.2 The frozen and the fine-tuned baselines share one chain. Frozen: the encoder runs once and its token vectors are stored. Fine-tune: the encoder trains with the head. White: input. Yellow: the blocks of the baselines. 12

7.1 **The per-topic view of the dashboard**, final model, validation data, over the 291 scored topics. Left: AP of each topic against its number of training articles, log scale, 1 dot per topic, coloured by level. Right: the topics ranked by AP inside each level, with the macro AP line. 26

A.1 The Graphormer label-attention baseline. The topic side on the left and the article on the right meet in the label attention, where each topic reads the article for itself. White: inputs. Yellow: the blocks of this baseline. 34

List of Tables

1.1 Targets and constraints of the model. Quality is measured against the LLM topics, so the LLM topics set a ceiling for the model. 2

2.1 The 2 encoders compared in this report. Sizes in parameters; input length in tokens. 5

3.1 The 3 data sets. The easy data set is mine; the gold training set and the real test set are the team's. 7

4.1 The blocks of the model in the order of Figure 4.1, where each one comes from, and what it does. 11
4.2 The baselines, in the order they were tried on the easy data set. All share the split, the topic set, the loss family and the metrics of the HGCLR runs. 11
4.3 Settings of the blocks in the run configuration. The HGCLR rows follow the paper and its reference code; the other rows are our blocks. 13
4.4 The 5 coherence settings. The rule in training sits inside the classification loss; the rule at inference runs on $p = \sigma(s)$ before the threshold. 17
4.5 The 2 rules on one chain root $\rightarrow$ A $\rightarrow$ a. The avg-max rule averages (A, a), then (root, a), then (root, A). 17

5.1 The training environment. Table A.1 gives the training settings. 19

6.1 **The 4 groups of runs.** Scores compare within a group, not across groups. Groups 3 and 4 share their validation split. 20
6.2 **Group 1: easy data set, old platform.** Baselines, validation split. One seed, or the ensemble of several seeds. Weighted BCE and no coherence rule unless stated. 21
6.3 **Group 1: easy data set, old platform.** The ASL sweep on the frozen baseline, 1 seed. The weighted BCE row comes from a later run of the same baseline. Macro AP is the sort key of Chapter 6. 21
6.4 **Group 1: easy data set, old platform.** Coherence rules. Each block differs in the rule alone. The last 3 columns give macro F1 with no rule, with the max rule and with the avg-max rule applied to the saved scores; a run trained with a rule in the loss has the same value in all 3. 22
6.5 **Group 1: easy data set, old platform.** HGCLR, 1 seed each, weighted BCE unless stated. Frozen rows: the positive view comes from a mask that keeps the stated share of tokens; a guided mask keeps the tokens closest to the names of the article's topics. Fine-tune rows: the positive view comes from the graph-guided generator of Section 4.3.2. The positive views of a frozen run are encoded once and cached; the 2 rows with no mask differ only in how often the cache is rebuilt, which with no mask changes the random stream alone. The plain fine-tune row repeats Table 6.2. 23
6.6 **Groups 3 and 4: pruned gold set, rebuilt pipeline**, full corpus. Compare rows within a group only. One seed unless stated. 23

6.7 The evidence for the rules. A: each rule applied after training to the saved scores of the 58 runs of the old platform trained without a rule, macro F1 on the validation data, **groups 1 and 2**. B: the rule inside the training loss, macro AP on the validation data; **each pair sits inside 1 group** of Table 6.1; a dash marks a pair that was not run. *The 2 runs differ in their soft values and article-to-topic attention as well. 24
6.8 The final design with 3 seeds and their ensemble. **Group 4**, full corpus. 24

7.1 The 32 topics the model learns and the metrics skip, by the reason found for each. 25
7.2 The final model by level. Macro AP, and precision and recall over all article-topic pairs at the threshold of 0.5. 26
7.3 4 causes of a disagreement on the validation data, 1 article each. The LLM topics and the model's topics at level 1 and 2, where they differ. 27
7.4 A validation article where the model and the LLM topics disagree. *Une retraitée d'Ajaccio tricote une écharpe de 21 kilomètres, chantier textile inédit en Corse*: a 72-year-old retired woman finishes a scarf 21 km long; the body stops at a paywall after 2 sentences, as the generator cuts a share of the articles on purpose, the way scraped news arrives. Votes: LLMs out of 3 that gave the topic; 0 means no LLM did. 27
7.5 Blind LLM judge on validation disagreements, earlier run of the same design: 150 random validation articles, 1,066 disputed topics. 1 judge, name of the judge LLM to fill in, no human check. The rest of each row is undecided. 27
7.6 Share of articles that carry a level 1 topic, in the generated validation data and in the real test set, and the model's results on the real test set at the threshold of 0.5. Topics with the largest change. 28

A.1 Training settings of the final model. 36

Chapter 1

Introduction

1.1 Abstrakt and AI-assisted press monitoring

Abstrakt is a French start-up that monitors the press for its customers with the help of AI. The press publishes more than a person can read, so the AI picks the articles that matter for each customer, and a person reads what the AI kept. **AI narrows the reading and people decide.** For a model this means that recall comes first, since a customer does not see a missed article, and that precision must stay high, since a person pays for each wrong article with reading time.

1.2 The mission: a topic tag on each article

Each article in the service carries topic tags. The tags come from the IPTC Media Topics list, the standard of the news industry. The topics form a tree. Level 1 has 17 broad topics, for example *politics* and *government* or *sport*. Level 2 has 120 topics and level 3 has 515. The project stops at level 3. Many articles have several topics. A topic implies its parent topics up to the root.

The product uses the tags in 2 places. With a level 3 tag on each article, a customer can filter the articles on one precise topic, as in a traditional press review. The level 2 tags serve as filters in the search bar.

A large language model (LLM) can read an article and classify its topics from the list. The team started this way, and the cost was the problem: **classifying each article with an LLM costs too much** at the scale of the service. My mission was to find a better way. A small model is trained once, then runs on 1 GPU and tags each article at a very low cost. A small model learns from examples, and the examples were missing. So the LLMs keep one job, writing the training data. Several LLMs vote on the topics of each training article, and the model learns from the vote. A vote of several LLMs is more reliable than 1 LLM, so the model can tag better than the single LLM the team started with.

3 more facts guided the design. Copyright forbids training the model on real articles. The LLM therefore writes training articles for each topic, and the team keeps real articles for the test data. The tree has 652 topics down to level 3. The team cut the niche topics and the topics with no business use, and of the topics that remain, many appear in a few articles. These rare topics drive the macro scores. The model runs on one small GPU in production. One encoder with a small head on top fits this GPU, so I kept that design and did not try one model per node of the tree.

The team chose the design at the start: a pre-trained text encoder, EuroBERT, and a head on top that outputs one score per topic. My work during the internship was to train the best model I could within this design. In practice this meant **choosing the encoder and its size, the way to train it, the pooling and the head, the loss, and a rule that keeps the scores coherent along the tree**; the one method that teaches the encoder the tree itself,

HGCLR, had to be rebuilt for EuroBERT. Chapter 4 presents these choices and Chapter 6 compares them.

1.3 Objectives and constraints

Table 1.1 lists the targets and the constraints of the model. **The quality targets apply per level and loosen with depth**, since the LLM topics are less reliable there. The final model meets them on the validation data, Table 7.2. The real test set scores lower, in part because its LLM topics are not fully clean, in part because its mix of topics differs, Chapter 7. The daily volume grows 10 times over the year. Inference runs on one small GPU, and a 210m or a 610m encoder fits it. The team spends the LLM budget on training data.

Table 1.1: Targets and constraints of the model. Quality is measured against the LLM topics, so the LLM topics set a ceiling for the model.

Item	Scope	Target	Note
Recall, precision	Level 1 (17 topics)	$R \geq 0.90$, $P \geq 0.85$	on the validation data
Recall, precision	Level 2 (106 topics)	$R \geq 0.85$, $P \geq 0.70$	
Recall, precision	Level 3 (168 topics)	$R \geq 0.80$, $P \geq 0.60$	looser as the topic count grows
Throughput	production	20k articles/day by September	200k/day by January
Hardware	production	one small GPU	a 210m or a 610m encoder fits
Languages	input	French first, multilingual later	

1.4 Contributions of the internship

The work has 4 parts.

1. **Training data.** The easy data set (v2, Chapter 3): a random sample of the titles, the rare topics folded into their parent topic, and 1 article per title written and tagged by a small LLM. It carried the model work for 2 months. The gold data set and its real test set (Chapter 3), built with the team in September: several LLM runs vote on each topic, and rules clean the data and drop the rarest topics.

2. **A model that knows the tree.** HGCLR is a training method that teaches the encoder the tree. For each article it keeps the words that its topics point to, and it trains the encoder to give the same output for the full article and for the kept words. I rebuilt HGCLR with EuroBERT in place of BERT and with 4 changes: attention pooling in place of a single token or a simple pooling, a deeper head, the ASL loss, and an attention block from the article to the topics before the pooling. I compared 3 ways to make the output scores coherent, which means a parent score at least as high as each child score. One of them is a new averaging rule that moves a parent and its child toward each other; it costs about as much as the max rule of C-HMCNN, keeps the scores coherent in the same way, and lowered the scores in fewer runs (Chapter 4).

3. **Experiments on the easy data set.** I built one platform to compare each choice of the project on the same footing: encoders, model structures, losses, optimizer settings and coherence rules, each with several seeds, and every run stored with its commit hash and its metrics (Chapter 6).

4. **Analysis tools.** A dashboard that shows the scores of a run per level and per topic, and lists the topics and the articles the model gets wrong, so that an error can be traced to a topic, an article or a label (Chapter 7).

A short side study, Chapter 9, compares open rerankers for thematic search, a nearby need of the product.

1.5 Outline of the report

Chapter 2 gives the background: the topic tree, the task and the methods it borrows from, the encoders, the losses and the metrics. Chapter 3 describes the 3 data sets. Chapter 4 presents the model, from the baselines to HGCLR, my changes and the coherence rules. Chapter 5 states the environment of the runs. Chapter 6 reports the results, first on the easy data set and then on the gold training set and the real test set. Chapter 7 analyses the errors. Chapter 8 lists the limits of the work. Chapter 9 reports the reranker study, and Chapter 10 ends with what is new in the work and what I learned.

Chapter 2

Background

2.1 The IPTC Media Topics tree

IPTC Media Topics [7] is the topic list of the news industry, kept by the International Press Telecommunications Council. Each topic has a code, a name and a one-line definition in many languages. The topics form a tree: each topic has one parent, and the tree goes 6 levels deep. The project uses the first 3 levels: 17 topics on level 1, 120 on level 2 and 515 on level 3. The level 1 topics have no common root, so the tree is a forest of 17 trees. A press article gets a set of topics, and a topic implies its parent topics up to level 1: an article about a football match is about *football*, *competition* and *sport* at once.

2.2 Hierarchical multi-label classification

The task is multi-label: an article can have several topics, so the model gives one score per topic and a threshold turns each score into a yes or a no. The task is hierarchical too: the set of topics of an article contains the parents of each of its topics, so a child topic never comes without its parent. A set of predictions is coherent when each parent scores at least as high as each of its children; a threshold on coherent scores then keeps the tree intact.

A flat model gives one score per topic and knows nothing of the tree. It can say *football* and deny *sport*. Hierarchy-aware models bring the tree in at 1 of 2 places. Some put it into the model, so that the vectors of the topics or of the article carry the tree. Others put it at the output, with a rule that makes the scores coherent. This report tries both places and compares them, alone and together: HGCLR puts the tree into the encoder, and a coherence rule makes the scores coherent, at inference and, in one variant, in training as well.

HiAGM [15] is an earlier model that puts the tree into the model: a graph network turns the tree into one vector per topic, and each topic vector attends over the text to get its own view of it. The Graphormer label-attention baseline of Chapter 4 follows this design. HGCLR [13] uses the topic vectors in training alone, to build the positive view, and its paper reports higher scores than HiAGM on the 3 usual benchmarks.

HGCLR works as follows. **A graph encoder turns the topic tree into one vector per topic**. For each training article, a generator uses these vectors and the article's topics to keep the tokens that speak for the topics and to mask the rest; this masked copy is the positive view of the article. The encoder reads the article and its positive view, and a contrastive loss pulls their 2 vectors together while pushing the other articles of the batch away. The encoder thus learns which tokens carry the topics and keeps this knowledge at inference, where the model is a plain encoder with a head. Chapter 4 gives the blocks and the losses.

2.3 Encoders for French press

An encoder is a transformer that reads a whole text and gives one vector per token, where each vector carries the context of the token. Encoders of the BERT family are pre-trained on large text with masked tokens to fill in, then fine-tuned on the task at hand. Most of the articles here are French press articles of some length, and the product must handle other languages too, so the encoder must read French well, handle several languages and accept long inputs. Table 2.1 lists the 2 encoders I compared.

Table 2.1: The 2 encoders compared in this report. Sizes in parameters; input length in tokens.

Encoder	Languages	Sizes	Trained for	Input length
EuroBERT	European languages	210m, 610m, 2.1B	masked tokens	8,192
multilingual-e5-base	about 100 languages	280m	retrieval	512

EuroBERT [1] is a recent encoder for European languages, in 3 sizes, with the recipe of recent decoder models: rotary positions, no bias, and a long input window. Its tokenizer adds an end token and no start token, a detail that matters for the pooling in Chapter 4. multilingual-e5 [12] is an older multilingual encoder trained for retrieval, so its vectors are made for comparing texts, and it reads short inputs.

2.4 Coherent predictions: C-HMCNN and the max rule

C-HMCNN [4] makes the output of any multi-label model coherent with one rule, the max rule. Let p_j be the score of topic j and D_j the set made of j and all its descendants. The output score of j is

$$\mathrm{MCM}_j = \max_{k \in D_j} p_k. \tag{2.1}$$

A parent then scores at least as high as each child, and a threshold on these scores respects the tree.

The rule alone has a flaw in training: a child with a wrong high score lifts its parent, and the loss then pushes the parent down instead of the child. C-HMCNN therefore trains with a loss where, for a topic the article has, the maximum runs over the descendants it has too:

$$L_j = -y_j \ln\Big(\max_{k \in D_j} y_k\, p_k\Big) - (1 - y_j) \ln\big(1 - \mathrm{MCM}_j\big). \tag{2.2}$$

The gradient then reaches the topic that caused the error. Chapter 4 uses this rule and derives from it the avg-max rule of the final model.

2.5 Losses for imbalanced multi-label data

The usual loss for multi-label classification is binary cross-entropy, one yes-or-no loss per topic. With hundreds of topics and a few topics per article, almost all the terms are negatives, and the model learns to say no. The first fix weights the positive term of each topic by how rare the topic is, with a cap on the weight. The second is the asymmetric loss, ASL [11], and Figure 2.1 shows its idea. The top plot draws the loss on one negative against the score the model gives it. Cross-entropy keeps a loss on each negative down to a score of 0, so the many easy negatives add up and drive the training. Focal loss scales this loss down for a negative with a low score. ASL goes one step further and shifts the curve to the right by a margin m, so a negative with a score below m costs nothing. The positives keep the full loss. The bottom plots show the result on MS-COCO, where p_t is the model's confidence in the right answer. With

cross-entropy the negatives reach a high confidence within the first steps while the positives stay below. With ASL the 2 curves meet. ASL has 3 settings, one exponent per side and the margin, and with all 3 at 0 it is binary cross-entropy again. Chapter 4 states the values used.

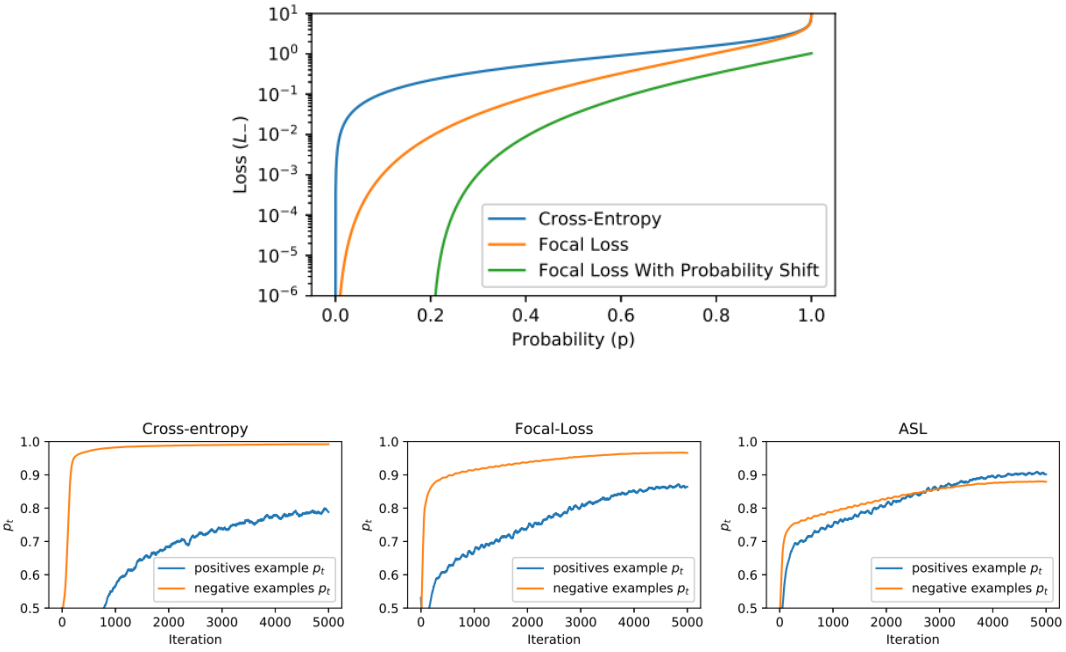


Figure 2.1: The idea of ASL, 2 figures from Ridnik et al. [11]. Top: the loss on a negative against its score, for cross-entropy, focal loss [9], and focal loss with the margin m. Bottom: the mean confidence in the right answer, p_t, of the positives and of the negatives during training on MS-COCO, with cross-entropy, focal loss and ASL.

2.6 Evaluation

The model gives 1 score per topic. With a threshold each score becomes a yes or a no, and precision, recall and F1 follow. Without a threshold, average precision (AP) rewards a model that ranks the right topics first. Micro versions count over all articles at once, so frequent topics dominate; macro versions average over the topics, so each topic counts the same. **Macro AP is the headline number of this report.** 1 more number serves the tree: the break value B, 0 when the scores respect the tree. Appendix B gives the formulas.

Chapter 3

Data: 3 data sets

3.1 Starting point

At the start of the internship no training data existed for the task: no article with its 3 levels of topics. The team had 2 things. The first was a set of titles from its thematic search work, each title with a level 1 and a level 2 topic and no article text. The second was a scrape of real French press articles, with no topics yet. Copyright keeps real articles out of training, so this scrape could serve as test data alone. 3 data sets followed, Table 3.1: an easy data set to try the methods, a gold training set, and a real test set.

Table 3.1: The 3 data sets. The easy data set is mine; the gold training set and the real test set are the team's.

	Easy data set	Gold training set	Real test set
Built by	me, August	the team, September; pruned 21 September	the team, September; trimmed 5 October
Text	LLM-written from a title	LLM-written from an LLM-written seed title	real French press, scraped in 2026
Writer LLM	gpt-5.4-nano	**gpt-5-mini**	none
Topics given by	the same LLM, 1 vote	3 LLMs, all votes kept	the same 3 LLMs
Languages	French	2/3 French, 1/3 English	French
Articles	10,798	58,527 (59,495 before pruning)	9,701 (10,000 before trimming)
Topic list	294 distinct; 186 kept by the model	17 level 1, 120 level 2, 192 kept level 3	the same list
Use	compare methods, Sections 6.1 to 6.4	train the model, Section 6.5	measure the model, never train

3.2 The easy data set

My first design, in the first week of August, had an LLM read real articles of one topic, write a profile of them, and generate articles from that profile. A profile is an average, and the model needs many kinds of writing per topic, so I dropped it within the week. The second design, which produced the data set within the same week, starts from the titles instead. The team had generated these titles earlier as niche titles: unusual kinds of news within each topic, not the average news of the topic, so that a model learns the edges of a topic and not its centre alone. The design runs in 4 steps. First I picked a sample of the titles, spread over the level 1 and level 2 topics, with a cap per pair of topics so that no topic floods the sample. Second, a small LLM wrote one press article from each title. Third, the same LLM read the article and tagged it with

a tree of topics down to level 3; a tag outside the tree, or under the wrong parent, went back to the LLM, and after a few tries the invalid part was cut. I dropped the articles whose level 1 topic did not match the title's. Fourth, I folded the long tail: a topic with too few articles went into its parent topic, so that the data set holds topics the model can learn. One seed fixes the split into training and validation data, and the model's topic list drops the topics that appear in less than 1 % of the training articles.

3.3 The gold training set

The gold training set is the team's work, built in September; I took part in the design discussions. It keeps the idea of the easy data set, LLM-written articles, and fixes its 2 weaknesses: 1 LLM as both writer and judge, and topics with no articles. The team cut the tree to the topics the product needs. For each kept topic 1 LLM wrote seed titles and a stronger LLM wrote 1 article per title, in French and in English, in several press formats and lengths. The topics of an article come from an LLM in 2 stages: it first picks the level 1 topics, and then receives the subtrees of those topics alone to pick the deeper ones, which keeps the prompt short. Several LLMs tag each article this way; a stronger LLM settles their disagreements.

The voters give too many topics, so from the first version on a judge LLM answered yes or no for each level 1 topic of an article. **A later pass pruned the topic tree alone.** The article count of Table 3.1 fell with it, since an article left without a topic left the set. The set before pruning stays on the Hub.

3.4 The real test set

The real test set is real French press scraped in 2026, tagged the same way as the gold set: the same LLMs, the same votes, the same judge. It measures the model and takes no part in training. The team drew the sample with about the same number of articles per level 1 topic. The new version of the set removes the pages with no article behind a paywall or a cookie wall. Its topics are LLM votes as well, since no human-tagged set exists yet.

3.5 Purpose and limits

Each data set has one job.

- **The easy data set explores the effect of each method and each change**, Sections 6.1 to 6.4. It is easy in 2 ways: the LLM that wrote an article tagged it as well, so the topics fit the text, and the rare topics are gone.
- **The gold training set trains the model for the product**, with the full topic list.
- **The real test set measures the model on real articles.**

2 limits hold for the 3 data sets. The topics come from LLMs, not from people, so a score measures how far the model agrees with the LLMs. And the generated articles differ from real press in which topics appear and in their shares, a gap that Chapter 7 measures.

Chapter 4

Method

4.1 Model overview

The model takes 2 inputs, shown on the top row of Figure 4.1: the IPTC topic tree, and an article with its title and text. In training the article also brings its topics y. The topic side of the figure works on the tree. Each topic gets a vector: the EuroBERT embedding of its name plus a learned vector of its own. The IPTC topics form a forest: one tree per level 1 topic. A Graphormer then mixes the vectors within each tree, so that each topic vector carries the whole tree it belongs to. The result is G, one vector per topic. G is learned during training and does not change at inference.

The article side is the solid chain in the middle. The input is one string, the title after the word `title:` and the text after the word `article:`, so the encoder knows which part is which. The tokenizer cuts this string into tokens, and the embedding table turns them into vectors e. The encoder g_θ, a pre-trained EuroBERT, reads e and outputs one vector per token. The article-to-topic attention scores each token vector against all the topics in G and scales up the tokens that match a topic. A small feed-forward layer then transforms each token vector on its own. Attention pooling merges the token vectors into one vector z for the article. The head maps z to one score per topic. The avg-max rule makes the scores coherent: a parent topic scores at least as high as each of its children. Each topic whose score passes the threshold becomes a tag of the article. This chain is the whole model at inference.

Training adds the positive view. The positive-sample generator compares each token embedding with G and masks the tokens that are unimportant for the article's topics y. This masked copy $\hat{e}$ goes through the same chain as the article and gives a second vector $\hat{z}$ and a second set of scores. The training loss then asks 2 things: the scores of the article and of the positive view must match y, and z and $\hat{z}$ must be close. Section 4.3 gives the losses.

Table 4.1 sums up the blocks. **HGCLR [13] gives the frame**: the topic side, the generator and the contrastive loss, on top of a BERT encoder, here EuroBERT. Inside this frame I replaced 4 blocks with my own: the article-to-topic attention with its feed-forward layer, the pooling, the head and the classification loss. On the output side I added avg-max, a coherence rule derived from C-HMCNN [4]. The rest of this chapter presents these parts one by one, starting with the baselines I compared the model against.

4.2 Baselines

The work started on the easy data set with the simplest model: a frozen encoder and a head on top. This model is cheap enough to run many times, so I used it first to compare encoders, different BERT models and different sizes. The models that followed add one thing each: training the encoder, a cheaper way to train it, and a head that knows the tree. All of them share the

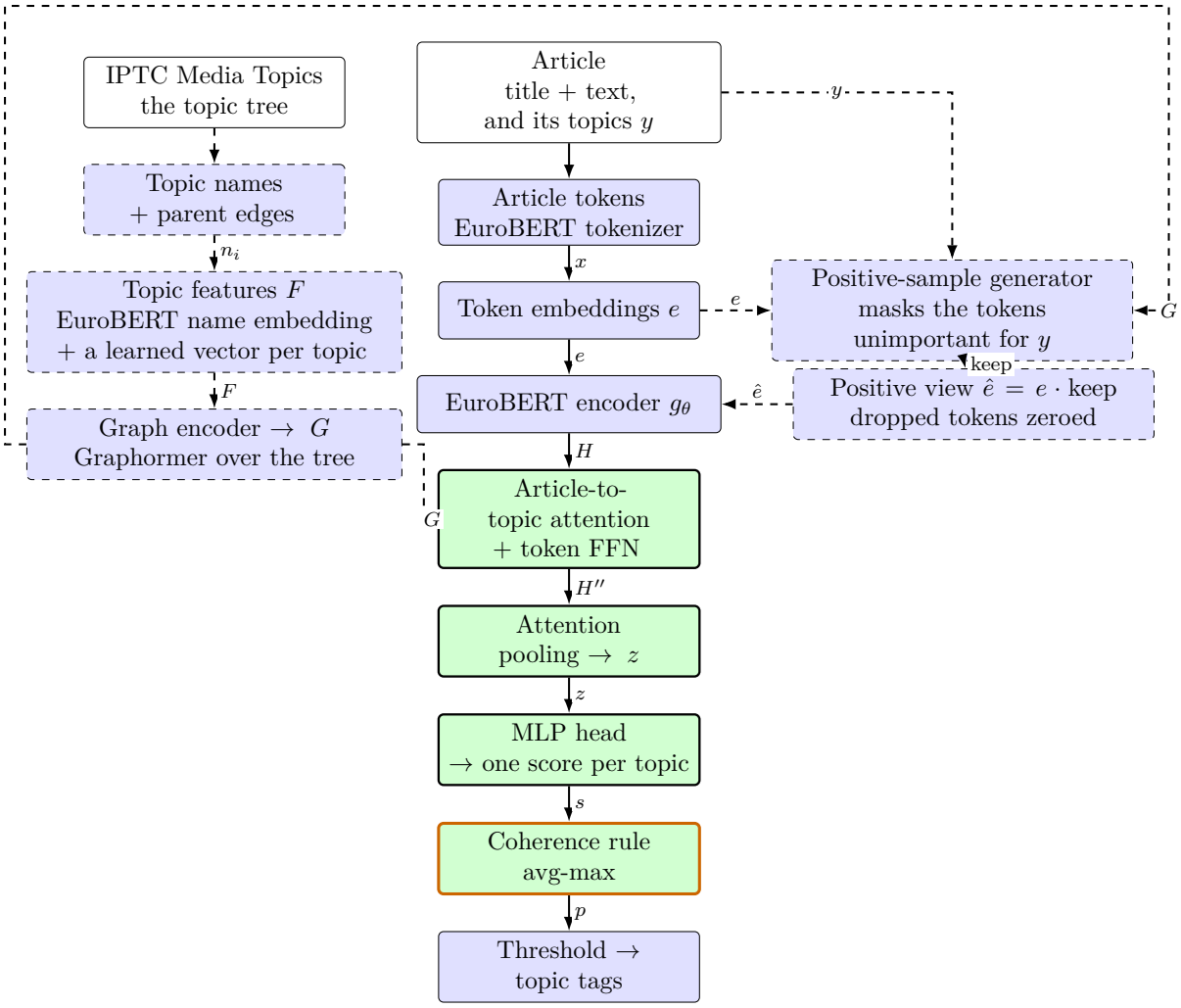


Figure 4.1: The model: the data flow from the inputs on the top row to the topic tags. White: inputs. Blue: HGCLR blocks. Green: our changes. Green with an orange border: avg-max, our variant of the C-HMCNN rule. Solid: the inference path, one pass per article. Dashed: training only. The letters on the arrows are the symbols of the equations in Section 4.3; the positive view $\hat{e}$ follows the same chain as the article and gives $\hat{z}$ and $\hat{s}$. The losses are in Section 4.3; the full graph with tensor shapes is in Appendix C.

Table 4.1: The blocks of the model in the order of Figure 4.1, where each one comes from, and what it does.

Block	Origin	What it does
Topic features F	HGCLR	one vector per topic: the EuroBERT embedding of the topic name plus a learned vector
Graph encoder G	HGCLR	a Graphormer mixes the topic vectors within each tree of the forest
Encoder g_θ	EuroBERT	a pretrained BERT; reads the article tokens and, in training, the positive view
Positive-sample generator	HGCLR	in training, keeps the tokens that match the article's topics y and masks the unimportant tokens
Article-to-topic attention + token FFN	ours	scores each token against all the topics in G and scales up the tokens that match a topic, then a small feed-forward layer on each token
Attention pooling	ours	merges the token vectors into one vector z per article
Head	ours	3 linear layers from z to one score per topic
Classification loss	ours	asymmetric loss (ASL) on the scores
Contrastive loss	HGCLR	pulls z and $\hat{z}$ together
Coherence rule	C-HMCNN, ours	avg-max: a parent topic scores at least as high as each child; our variant of the C-HMCNN rule

split, the topic set, the loss family and the metrics. Table 4.2 lists them; the results are in Chapter 6.

Table 4.2: The baselines, in the order they were tried on the easy data set. All share the split, the topic set, the loss family and the metrics of the HGCLR runs.

Step	Model	Encoder	What trains	Question it answers
1	Frozen + MLP head	EuroBERT 210m, 2.1B; multilingual-e5. Frozen, one pass, vectors stored	the head	which encoder, and how far a frozen encoder goes
2	Fine-tune	EuroBERT, all layers or the top layers trained	encoder and head	what training the encoder adds, and how much of it
3	LoRA	EuroBERT, weights fixed, low-rank adapters	adapters and head	fine-tune quality at a fraction of the memory
4	Graphormer label attention	EuroBERT, frozen or trained	Graphormer, topic attention, (encoder)	whether a head that knows the tree helps

4.2.1 Frozen encoder and an MLP head: choosing the encoder

The first model keeps the encoder as it is. It runs each article through the encoder once and pools the token vectors into one vector per article. The pooling is a fixed rule: the mean and the maximum over the tokens, side by side. Figure 4.2 shows the chain. I store these vectors on disk, so the encoder never runs again. The head is a standard classifier, a small MLP trained on the stored vectors with binary cross-entropy. It is the baseline for everything that follows: the choice of encoder, the architectures, the learning rates and the schedules. Training the head takes minutes, so I can run many settings and repeat each one with several seeds.

Because the encoder runs once, **this model is the right place to compare encoders**. I compared 2 models, EuroBERT and multilingual-e5, and 2 sizes of EuroBERT, 210m and 2.1B.

The encoders differ in the length they accept, the prefix their input needs and the pooling their weights were trained for, so I wrote these facts into one table in the code, and one training loop reads it. EuroBERT-2.1B scored higher, but a full fine-tune of it costs far more time and memory, so EuroBERT-210m became the encoder for the main runs. If a change made no clear difference on 210m, I tried it again on the bigger model. Chapter 6 gives the numbers.

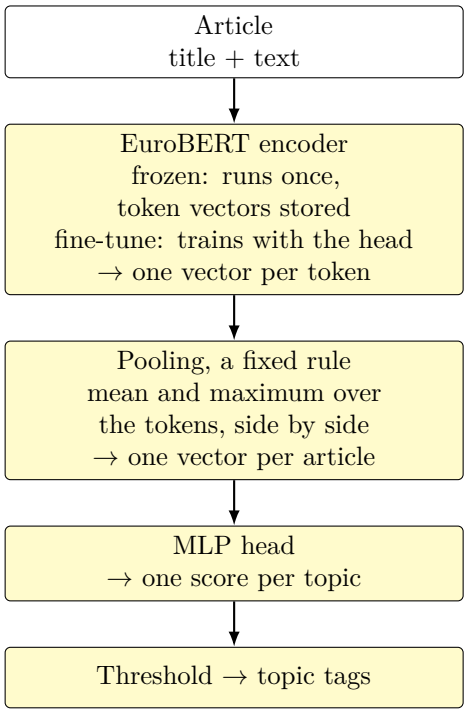


Figure 4.2: The frozen and the fine-tuned baselines share one chain. Frozen: the encoder runs once and its token vectors are stored. Fine-tune: the encoder trains with the head. White: input. Yellow: the blocks of the baselines.

4.2.2 Training the encoder: warm-up and learning rates

The second model trains the encoder as well as the head, and it beats the frozen model. The head starts random, so its first gradients are noise, and noise pushed into a pretrained encoder distorts the features it learned. 2 standard settings prevent it: the encoder stays frozen for the first epochs while the head trains, which I call the warm-up, and then trains with a lower learning rate than the head. This full fine-tune model costs more than the frozen one in 2 ways. The optimizer, AdamW, keeps 2 running averages for each weight of the encoder, so the weights, their gradients and the 2 averages take about 4 times the memory of the weights alone. And each run saves the whole encoder to disk, where the frozen model saves a small head.

2 cheaper options, training the top layers alone and LoRA [5], sit between the frozen encoder and the full fine-tune; Appendix A.2 describes them. **Neither made it into the final model, which trains the whole encoder.**

Chapter 6 compares the options.

4.2.3 Graphormer label attention

The third model is the first one that sees the tree: a Graphormer runs attention between the topics, and each topic then reads the article through the label attention of HiAGM-LA [15]. Appendix A.1 describes it. It scores about the same as the fine-tuned baseline, Section 6.1.

4.2.4 Several seeds: the spread and the ensemble

A small gain may be noise. The seed sets the random start of the head, the dropout and the batch order, and 2 seeds of the same setting can differ by more than the gain. So I ran each setting with several seeds: more for the cheap models, fewer for the models where each seed is a full fine-tune. A change counts as real when it beats the spread of the seeds. The seeds give an ensemble too: the mean of their probabilities, which scores much higher than any single seed. For the frozen model this is free, since each seed is one more head on the same stored vectors. Chapter 6 reports both.

4.3 The model block by block

At inference the model sees the article alone. The topic tree can help it then in one way: **the encoder must have learned the tree during training**. Hierarchy-guided contrastive learning (HGCLR) [13] does this by making a positive view of each training article: a copy that keeps the tokens about the article's topics and masks the rest. A generator reads the article's topics y and chooses these tokens. The encoder reads the article and this positive view with the same weights, and a contrastive loss pulls together the 2 output vectors, one for the article and one for its positive view. The encoder learns to find the topic-bearing tokens on its own, and at inference it does so without y. This section walks through the blocks of Figure 4.1 in the order of the data flow. The topic side, the generator and the contrastive loss come from HGCLR; the article-to-topic attention, the pooling, the head and the classification loss are mine. Table 4.3 lists the settings of the blocks.

Table 4.3: Settings of the blocks in the run configuration. The HGCLR rows follow the paper and its reference code; the other rows are our blocks.

Block	Origin	Setting	Value
Graph encoder	HGCLR	Graphormer layers	1
		attention heads	12
Generator	HGCLR	attention heads	8
		keep threshold γ	0.02
		Gumbel temperature τ	1
Article-to-topic attention	ours	attention heads	8
		token FFN hidden width	768
Attention pooling	ours	heads	12
		MLP hidden width	128
Head	ours	hidden width	1024
		layers	3
		dropout	0.2
Classification loss (ASL)	ours	γ_+	0
		γ_-	1
		margin m	0.05
Contrastive loss	HGCLR	temperature τ_{con}	1
		weight λ	0.1

4.3.1 Topic features and the graph encoder

Each topic i starts as a vector f_i with 2 parts. The first part is the mean of the EuroBERT token embeddings of the topic name, so the graph encoder starts with basic information on each topic. The second part is a learned vector E_i of the topic's own, which training moves as it

needs to tell topics apart:

$$f_i = E_i + \text{name}_i, \qquad \text{name}_i = \frac{1}{|n_i|} \sum_{w \in n_i} \text{embed}(w), \tag{4.1}$$

where n_i is the set of tokens in the name of topic i, w runs over these tokens, embed(w) is the EuroBERT embedding of token w, and f_i is row i of F. I set the start scale of E to the scale of the token embeddings.

The graph encoder is one Graphormer layer [14], as in HGCLR: attention among the topics, where the attention score of a pair of topics gets a learned bias from their distance in the tree and from the edges on the path between them:

$$a_{ij} = \frac{(W_Q f_i) \cdot (W_K f_j)}{\sqrt{d}} + b_{\text{dist}(i,j)} + c_{ij}, \tag{4.2}$$

with b one learned value per distance and c_{ij} the mean of a learned value per edge on the path from i to j. A softmax over j and a weighted sum of the f_j follow, as in any attention layer. **The output is G, one vector per topic that mixes in all the other topics of its tree**, weighted by distance; the tree knowledge itself sits in the learned biases and topic vectors, and G is recomputed from them at each step. In HGCLR, G serves the generator alone. In our model the article-to-topic attention of Section 4.3.3 reads G too, at inference as well.

4.3.2 Positive-sample generator and the 2 views

The generator decides which tokens of a training article survive into the positive view, as in HGCLR. It works on the token embeddings e_i, before the encoder. The equations follow the paper. Each token x_i is a query and each topic j, with its row G_j, is a key:

$$q_i = e_i W_Q, \qquad k_j = G_j W_K, \qquad A_{ij} = \frac{q_i k_j^\top}{\sqrt{d_h}}. \tag{4.3}$$

A softmax over the topics would give the probability that token x_i belongs to topic j. **HGCLR replaces it with a Gumbel-softmax** [8]**, which adds noise and keeps the sampling differentiable**:

$$P_{ij} = \text{gumbel_softmax}(A_{i1}, A_{i2}, \ldots, A_{ik})_j. \tag{4.4}$$

The sum over the article's topics y gives the probability of the token under its topic set, and a threshold γ builds the positive view:

$$P_i = \sum_{j \in y} P_{ij}, \qquad \hat{e}_i = \begin{cases} e_i & \text{if } P_i > \gamma \\ 0 & \text{else} \end{cases}. \tag{4.5}$$

The threshold has no gradient, so **the generator passes the hard choice forward and the gradient of P_i backward**. In HGCLR this is the one path that trains the topic side. The noise makes the choice random, so the same article gets a different positive view at each epoch.

In the positive view $\hat{e}$, **a kept token keeps its embedding and a dropped token becomes a zero vector at its position**. I run the encoder once: the article and its positive view go into one batch of $2B$ rows, which works because EuroBERT accepts embeddings as its input. From the encoder on, the 2 views go through the same blocks with the same weights, and come out as the vectors z and $\hat{z}$ and the scores s and $\hat{s}$.

4.3.3 Article-to-topic attention and token FFN

The generator highlights the tokens that carry the article's topics, but it reads y to do so, and y exists in training alone. **I added a block that does the same at inference, where there is no y: the article-to-topic attention**. It sits after the encoder and scores each token state H_t, the encoder's output for token t, against the graph-aware topic features G of the graph encoder, with the same attention as the generator:

$$q_t = H_t W_Q, \qquad k_j = G_j W_K, \qquad S_{tj} = \sigma\Big(\frac{q_t k_j^\top}{\sqrt{d_h}}\Big), \tag{4.6}$$

averaged over the heads. **A sigmoid replaces the softmax, and the maximum over all topics replaces the sum over** y, so the block runs without y:

$$r_t = \max_j S_{tj}, \qquad H'_t = H_t + \alpha\, r_t\, H_t. \tag{4.7}$$

The block drops no token; **it scales up a token that matches a topic**, with one learned scalar α that starts at 0. A feed-forward layer on each token follows, with the same zero start:

$$H''_t = H'_t + W_2\,\mathrm{GELU}\big(W_1\,\mathrm{RMSNorm}(H'_t)\big), \qquad W_2 = 0 \text{ at the start.} \tag{4.8}$$

Both blocks begin as the identity, so the model starts as the one without them.

4.3.4 Attention pooling

HGCLR takes the vector of the first token as the article vector, the usual BERT choice. **EuroBERT has no such token**: its tokenizer adds an end token and nothing in front. The frozen baseline of Section 4.2.1 pooled with a fixed rule, the mean and the maximum over the tokens. The final model pools with attention, the gated attention pooling [6] of the torchmil library [2]: a small MLP gives each token an attention value, a softmax over the tokens normalises the values, and the article vector is the weighted sum of the token states:

$$\mathbf{f} = \big(\tanh(H''\mathbf{W}_1) \odot \mathrm{sigm}(H''\mathbf{W}_2)\big)\,\mathbf{w}, \qquad z = H''^\top \mathrm{Softmax}(\mathbf{f}) = \sum_{t=1}^{T} s_t\, H''_t, \tag{4.9}$$

where H'' holds one token state H''_t per row, $\mathbf{f} \in \mathbb{R}^T$ are the attention values, s_t is the normalised score of token t, and $\mathbf{W}_1$, $\mathbf{W}_2$ and $\mathbf{w}$ are the weights of the MLP. With several heads, $\mathbf{w}$ has one column per head, and **each head pools its own slice of H'' with its own scores**. The softmax skips the padding tokens. The article vector is a weighted mean of the token states, where the model chooses the weights and each head can look at different tokens. This block gave the largest single gain of my 4 changes on the easy data set, on the frozen baseline. A dropout on z follows, and z feeds both the head and the contrastive loss.

4.3.5 Head

HGCLR maps z to the scores with one linear layer. **Our head is a small MLP of 3 linear layers**, with a ReLU and a dropout after the first 2:

$$s = W_3\,\mathrm{ReLU}\big(W_2\,\mathrm{ReLU}(W_1 z)\big), \tag{4.10}$$

biases and dropouts left out. The last layer has no activation: s **is one raw score per topic**, and the sigmoid and the coherence rule come after. One head scores the article and the positive view. The frozen baseline used a head of the same shape, and I kept it in the final model.

4.3.6 Classification loss

The classification loss compares $p = \sigma(s)$ with the article's topics y, one term per topic. The baselines used binary cross-entropy with a capped positive weight per topic, against the imbalance of Section 2.5. The final model uses the asymmetric loss (ASL) [11]:

$$p^- = \max(p - m, 0), \qquad L^+ = (1-p)^{\gamma_+} \log p, \qquad L^- = (p^-)^{\gamma_-} \log(1 - p^-), \tag{4.11}$$

$$L^C = -\frac{1}{B} \sum_{\text{articles}} \sum_{j=1}^{L} \left[y_j L_j^+ + (1 - y_j) L_j^- \right]. \tag{4.12}$$

The margin m shifts the probability of a negative down and drops the easy negatives from the loss; the exponent γ_- dampens the ones that remain; γ_+ does the same on the positives. Figure 2.1 in Section 2.5 shows the effect. With all 3 settings at 0 the loss is binary cross-entropy. I swept them on the frozen baseline, without a detailed study. ASL scored higher than binary cross-entropy, and the setting with a light focusing on the negatives, a small margin and no focusing on the positives scored best; Table 4.3 gives the values, and Table 6.3 in Section 6.2 lists the sweep. The same loss runs on s and on $\hat{s}$.

4.3.7 Contrastive loss and the total loss

The contrastive loss pulls z and $\hat{z}$ together and pushes the other articles of the batch away. It is the NT-Xent loss of SimCLR [3], as in HGCLR. A small projection, 2 linear layers with a ReLU between them, maps each of the $2B$ vectors to a vector c and scales it to unit length, so that a dot product is a cosine. For each vector m the loss is a softmax over the other $2B - 1$ vectors, where the partner $\mu(m)$, the other view of the same article, must win:

$$L^{con} = -\frac{1}{2B} \sum_{m=1}^{2B} \log \frac{\exp\left(\text{sim}(c_m, c_{\mu(m)})/\tau_{con}\right)}{\sum_{i \neq m} \exp\left(\text{sim}(c_m, c_i)/\tau_{con}\right)}. \tag{4.13}$$

Each article has one positive pair and $2B - 2$ negatives in the batch. **The gradient of this loss flows back through both views into the encoder**. The positive view holds the tokens that speak for the topics and masks the rest, so to bring z close to $\hat{z}$ the encoder learns to build z from these tokens. The projection exists for this loss alone, and inference drops it.

The total training loss adds the classification loss on both views, with the same y, and the contrastive loss weighted by λ:

$$L = L^C(s, y) + L^C(\hat{s}, y) + \lambda L^{con}. \tag{4.14}$$

HGCLR uses binary cross-entropy for L^C; our model uses the asymmetric loss of Section 4.3.6. I kept λ at the value the paper reports for its 2 data sets, 0.1, and did not re-tune it after the change of loss. **At inference the generator, the positive view and the contrastive loss disappear**; the model is the solid chain of Figure 4.1.

4.4 Coherence rules

Section 2.2 defined a coherent prediction: each parent scores at least as high as each of its children. The head of Section 4.3.5 has no sense of the tree, so **a rule has to make the scores coherent**. One setting picks the rule, and with it where the rule runs: at inference alone, as a step after the sigmoid, or in training as well, inside the loss. Table 4.4 lists the 5 settings I compared; Chapter 6 gives their scores.

Table 4.4: The 5 coherence settings. The rule in training sits inside the classification loss; the rule at inference runs on $p = \sigma(s)$ before the threshold.

Setting	Rule in training	Rule at inference	Origin
none	none	none	flat baseline
hard_max	none	max	C-HMCNN, as a step after the sigmoid
hard_avg_max	none	avg-max	ours, as a step after the sigmoid
chmcnn	max	max	C-HMCNN
chmcnn_avg_max	avg-max	avg-max	ours; the final model

4.4.1 Rules at inference

A rule at inference takes the scores p of one article and returns coherent scores, and the threshold then runs on these. The max rule is the one of C-HMCNN, Equation 2.1: the score of a topic becomes the maximum over its subtree. It raises scores and lowers none, so it adds predictions and removes none; recall goes up and precision goes down, by construction.

The avg-max rule is my variant. Instead of lifting the parent to the child, it moves both to their mean: the parent goes up as much as the child goes down. It walks the tree from the deepest level up; inside a level it visits the topics in rising order of score; for each topic it climbs its ancestors, nearest first, and when an ancestor scores below the topic, both take their mean:

$$p_k \leftarrow \frac{p_k + p_j}{2}, \qquad p_j \leftarrow \frac{p_k + p_j}{2} \qquad \text{when } k \text{ is an ancestor of } j \text{ and } p_k < p_j. \tag{4.15}$$

In code, for one article:

```
for each level, from the deepest to the top:
    for each topic j of the level, in rising order of p[j]:
        for each ancestor k of j, nearest first:
            if p[k] < p[j]:
                p[k] = p[j] = (p[k] + p[j]) / 2
```

An ancestor rises and a topic falls, and no step does the reverse, so **after the pass each parent scores at least as high as each child, and the violation rate is 0 at any threshold**. Table 4.5 runs the rule on one chain. The max rule lifts the parents up to a wrong high child; the avg-max rule brings that child down as much as it lifts the parents, so the sum of the scores stays the same. The reason to trust the parent is the data: a parent has the articles of all its children, so it has more training data than any one child and scores with more reliability. The max rule hands the parent the child's score; the avg-max rule weighs the 2. The rule sorts the topics of a level by score so that the highest child meets its parent last; the parent arrives at that meeting lifted by the lower children, and so within one subtree the order of the topics holds.

Table 4.5: The 2 rules on one chain root → A → a. The avg-max rule averages (A, a), then (root, a), then (root, A).

	root	A	a
scores p	0.30	0.20	0.90
max rule	0.90	0.90	0.90
avg-max rule	0.4875	0.4875	0.425

4.4.2 Rules in training

A rule at inference repairs the scores and teaches the model no part of the tree. **C-HMCNN puts the rule inside the loss**. The loss of Section 4.3.6 then reads a corrected score p_j in

place of the raw score $h_j = \sigma(s_j)$:

$$p_j = y_j \max_{k \in D_j} y_k \, h_k \; + \; (1 - y_j) \max_{k \in D_j} h_k, \tag{4.16}$$

with D_j the subtree of topic j, as in Equation 2.1. **For a topic the article has, the maximum runs over the article's own topics. For a topic it has not, the maximum runs over the whole subtree.** A wrong high child thus raises the loss of its parent, and the gradient reaches the child. This is the setting chmcnn, Equation 2.2 with ASL in place of the cross-entropy. The setting chmcnn_avg_max puts the avg-max rule of Section 4.4.1 in place of each maximum. A soft value between 0 and 1 counts as a negative here.

A model trained with a rule in the loss predicts with the same rule at inference; its raw scores are not coherent on their own. Such a run converges slower than a flat run and needs more epochs. On the first gold set and on the full gold set, Table 6.7, the max rule in training lowered the scores. At inference the max rule lowered the scores in most runs, the avg-max rule in few. The final model therefore trains with the avg-max rule. Section 6.6 gathers the evidence.

4.4.3 The avg-max rule and its implementation

The max rule is simple to understand, and the sigmoid keeps the order of the scores, so the maximum can run on the logits, as 1 masked maximum over the whole batch on the GPU.

The avg-max rule cannot run on the logits, since the mean of 2 scores is not the sigmoid of the mean of their logits. It runs on the scores, per article, as the loop of Section 4.4.1. The final model records the averaged pairs on a tape in the forward pass and replays them in reverse in the backward pass. The loop runs on the CPU and takes about as long as the max rule on the GPU. Appendix A.3 gives the 3 versions and the gradient argument.

4.5 Training settings

The final model trains the whole encoder with AdamW [10], a warm-up and a cosine schedule, a batch of 16 articles on any GPU, the soft value for 1-vote topics, and early stopping on the validation loss. Appendix A.4 lists the values.

Chapter 5

Experimental setup

Training ran on rented cloud GPUs, 1 GPU per run, Table 5.1; a laptop held the code and its tests. Each run writes its configuration, the commit hash of the code and its metrics to a run store on the Hugging Face Hub, and an index page sorts the runs by their scores.

Table 5.1: The training environment. Table A.1 gives the training settings.

Item	Value
Cloud GPUs	NVIDIA T4 and L4 for the frozen baselines; A100, L40S, H100 and H200 for the fine-tunes and HGCLR
Precision	bf16 autocast, fp32 master weights
Attention kernel	flash attention 2 on Ampere and newer, SDPA otherwise
Software	Python, PyTorch 2.4 or later, transformers 5, numba for the avg-max rule
Run store	Hugging Face Hub, one folder per run with configuration, commit hash and metrics

Chapter 6

Results

The runs fall into 4 groups, Table 6.1, by their training data and their code. Scores compare within a group and not across groups, so each table names its group. Sections 6.1 to 6.4 use group 1, the later sections groups 2 to 4. A 1-seed row uses the same seed throughout; an ensemble averages the scores of its seeds.

Table 6.1: **The 4 groups of runs.** Scores compare within a group, not across groups. Groups 3 and 4 share their validation split.

Group	Training data	Code	Dates	Real test set
1	easy data set	old platform	Aug. to mid-Sept.	none
2	gold set, first version	old platform	14–16 Sept.	none in the index
3	gold set, pruned	rebuilt pipeline	23–30 Sept.	before trimming
4	gold set, pruned, 6 topics folded	rebuilt pipeline	from 14 Oct.	after trimming

6.1 Baselines on the easy data set: encoders, frozen and fine-tuned

Table 6.2 gathers the baselines of Section 4.2 on the easy data set, from the run index on the Hugging Face Hub. **The rows are not all on the same footing**: some have 1 seed and some an ensemble of seeds, and the frozen head changed its learning rate in September. The run index names the run behind each row.

3 results stand. **EuroBERT beats multilingual-e5** with a frozen encoder; the high recall of the e5 row comes from over-prediction. The large EuroBERT scores higher than the small one, at several times the time and memory; so the 210m is the base of the work, and the choice between the 2 sizes, trained or frozen, follows the requirements and the production budget. Training the encoder helps: the fine-tuned 210m beats the frozen 210m with 1 seed and with the ensemble. The attention pool on the frozen encoder recovers part of that gap, which is why Section 4.3.4 keeps it. The Graphormer label attention of Section 4.2.3 scores about the same as the fine-tuned baseline, within the spread of the seeds. LoRA has no row in the index, as Appendix A.2 says.

6.2 Losses and coherence rules on the easy data set

Table 6.3 lists the ASL sweep of Section 4.3.6 on the frozen baseline, 1 seed per run. **A light focusing on the negatives scores the highest macro AP**. A higher γ_- raises recall and lowers both F1 and macro AP. Focusing on the positives alone lowers recall. The adaptive runs,

Table 6.2: **Group 1: easy data set, old platform.** Baselines, validation split. One seed, or the ensemble of several seeds. Weighted BCE and no coherence rule unless stated.

Model	Encoder	Seeds	Macro AP	Micro AP	Macro F1	Macro recall
Frozen + head	EuroBERT 210m	1	0.563	0.710	0.530	0.672
Frozen + head	EuroBERT 210m	ens.	0.579	0.727	0.547	0.671
Frozen + head	EuroBERT 2.1B	1	0.581	0.722	0.548	0.633
Frozen + head	multilingual-e5	1	0.547	0.697	0.435	0.848
Frozen + attention pool	EuroBERT 210m	ens.	0.598	0.737	0.546	0.747
Fine-tune	EuroBERT 210m	1	0.606	0.753	0.560	0.683
Fine-tune	EuroBERT 210m	ens.	0.627	0.773	0.584	0.689
Fine-tune	EuroBERT 2.1B	1	0.612	0.752	0.504	0.835
Graphormer LA, fine-tune	EuroBERT 210m	ens.	0.627	0.760	0.601	0.636
Graphormer LA, frozen, max rule + ASL	EuroBERT 210m	ens.	0.632	0.775	0.588	0.646
Graphormer LA, frozen, max rule + ASL	EuroBERT 2.1B	ens.	0.624	0.767	0.581	0.634

where γ_- changes during training, score no better. The weighted BCE row comes from a later run, so it is a rough reference.

Table 6.3: **Group 1: easy data set, old platform.** The ASL sweep on the frozen baseline, 1 seed. The weighted BCE row comes from a later run of the same baseline. Macro AP is the sort key of Chapter 6.

Loss	γ_-	γ_+	m	Macro AP	Macro F1	Macro recall	Micro AP
BCE, no weight	0	0	0	0.571	0.507	0.449	0.726
BCE, capped positive weight	0	0	0	0.563	0.530	0.672	0.710
ASL	0	0	0.05	0.575	0.502	0.450	0.724
ASL	1	0	0.05	**0.581**	0.540	0.571	0.740
ASL	2	0	0.025	0.575	0.542	0.653	0.725
ASL	2	0	0.05	0.575	0.538	0.673	0.726
ASL	2	0	0.1	0.570	0.529	0.682	0.721
ASL	3	0	0.05	0.559	0.528	0.655	0.710
ASL	4	0	0.05	0.550	0.522	0.649	0.706
ASL	6	0	0.05	0.553	0.504	0.695	0.705
ASL	6	0	0.1	0.554	0.492	0.721	0.705
ASL	0	1	0	0.567	0.463	0.377	0.722
ASL, adaptive γ_-, gap 0.1	–	0	0.05	0.554	0.514	0.695	0.707
ASL, adaptive γ_-, gap 0.05	–	0	0.05	0.553	0.519	0.664	0.706
ASL, adaptive γ_-, gap 0.01	–	0	0.05	0.553	0.521	0.657	0.706
ASL, adaptive γ_-, gap 0.005	–	0	0.05	0.552	0.521	0.661	0.706

Table 6.4 compares the coherence rules of Section 4.4 in pairs that differ in the rule alone. **On the easy data set neither rule hurts the model**: with the max rule or the avg-max rule, in the loss or at inference, the scores stay within the noise of 1 seed, and a rule in the loss keeps the predictions coherent during training. The gold training set tells the 2 rules apart, Section 6.6.

6.3 HGCLR and our changes on the easy data set

Table 6.5 gives the HGCLR runs of Section 4.3 on the easy data set, 1 seed each. 2 findings shaped the final model.

The frozen encoder served to compare the masks. With the encoder frozen, a run takes minutes, so I used it to compare ways to build the positive view and to explore how many tokens the mask keeps. Table 6.5 shows the differences, all from 1 seed:

Table 6.4: **Group 1: easy data set, old platform.** Coherence rules. Each block differs in the rule alone. The last 3 columns give macro F1 with no rule, with the max rule and with the avg-max rule applied to the saved scores; a run trained with a rule in the loss has the same value in all 3.

Model	Rule in training	Seeds	Macro AP	Macro recall	F1 flat	F1 max	F1 avg-max
Frozen, ASL	none	ens.	0.592	0.582	0.552	0.552	0.550
Frozen, ASL	max (C-HMCNN)	ens.	0.593	0.580	0.557	0.557	0.557
Frozen, ASL	none	1	0.581	0.571	0.540	0.540	0.540
Frozen, ASL	avg-max	1	0.581	0.591	0.546	0.546	0.546
Fine-tune, BCE	none	ens.	0.627	0.689	0.584	0.583	0.586
Fine-tune, BCE	max (C-HMCNN)	ens.	0.630	0.689	0.584	0.584	0.584

- **The keep rate moves recall more than ranking.** A mask that keeps 90 % of the tokens, guided or random, raises macro recall by about 9 points and lowers macro F1 by about 3 against no mask, while macro AP drops by less than 0.5 point. Keeping 80 % lowers macro AP by about 1 point.
- **The guided mask edges out the random one.** At the same rate of 90 %, the mask guided by the topic names scores a little higher than the random mask on macro AP and recall.
- **No mask with the contrastive loss scores highest**, a little above the frozen baseline and above no mask without the contrastive loss. The run without the contrastive loss trained for more epochs, so this gap mixes 2 changes.

With a frozen encoder **the contrastive loss reaches the head and not the encoder** whose vectors it should shape, so all these gains stay small. The encoder has to train, so all later work fine-tunes it.

With a trained encoder HGCLR improves the metrics. The paper's schedule, a positive view at each step and all layers trained, scored below the plain fine-tune and trained much slower. 2 changes reversed this: training the top layers of the encoder alone, and building the positive view on 1 batch out of several, so that the other batches train like a plain fine-tune and skip the second pass through the encoder. The best run then beats the plain fine-tune of the same seed by about 1 point of macro AP.

The 2 blocks of Sections 4.3.3 and 4.3.4 came in with the final model on the gold training set, so they have no HGCLR row here.

6.4 Summary on the easy data set

The easy data set settled the design before the gold training set arrived: it shaped the final structure of the model and gave a good starting set of settings. EuroBERT 210m is the base encoder, trained with a lower learning rate and a warm-up; the loss is ASL with a light focusing on the negatives; a coherence rule removes the violations at no cost in ranking; HGCLR improves the metrics with the top layers trained and a sparse schedule of positive views; and the attention pool is worth its cost.

6.5 Runs on the gold training set

Table 6.6 gives the runs of the rebuilt pipeline on the full pruned gold set, in groups 3 and 4. Its scores are over the 291 scored topics; Table 7.2 gives the final model over all 323 topics as well.

Table 6.5: **Group 1: easy data set, old platform.** HGCLR, 1 seed each, weighted BCE unless stated. Frozen rows: the positive view comes from a mask that keeps the stated share of tokens; a guided mask keeps the tokens closest to the names of the article's topics. Fine-tune rows: the positive view comes from the graph-guided generator of Section 4.3.2. The positive views of a frozen run are encoded once and cached; the 2 rows with no mask differ only in how often the cache is rebuilt, which with no mask changes the random stream alone. The plain fine-tune row repeats Table 6.2.

Encoder	Positive view	Contrastive loss	Macro AP	Micro AP	Macro F1	Macro recall
frozen	none, the article itself; cache built once	yes	0.567	0.714	0.536	0.670
frozen	none, the article itself; cache rebuilt every 8 epochs	yes	0.565	0.713	0.537	0.653
frozen	guided mask, 90 %	yes	0.564	0.709	0.506	0.761
frozen	guided mask, 90 %, re-scored by the head	yes	0.564	0.708	0.500	0.768
frozen	random mask, 90 %	yes	0.563	0.707	0.503	0.756
frozen	none, the article itself	no	0.560	0.708	0.534	0.663
frozen	random mask, 80 %	no	0.548	0.694	0.509	0.709
frozen baseline, no HGCLR	–	–	0.563	0.710	0.530	0.672
fine-tune, all layers	generator, every step	yes	0.595	0.742	0.556	0.675
fine-tune, top 2 layers	generator, 1 batch in 10	yes	0.597	0.737	0.550	0.697
fine-tune, top 2, adaptive ASL	generator, 1 batch in 10	yes	0.604	0.755	0.455	0.858
fine-tune, top 6 layers	generator, 1 batch in 4	yes	0.613	0.756	0.571	0.677
fine-tune, top 6 layers	generator, 1 batch in 6	yes	0.616	0.757	0.574	0.680
plain fine-tune, no HGCLR	–	–	0.606	0.753	0.560	0.683

- **The HGCLR design beats the base**, group 4. The base fine-tunes the same EuroBERT 210m with the article-to-topic attention, the attention pool and the head, on the same data, input length and schedule, and leaves out 5 parts: the graph encoder, the positive view with its contrastive loss, the token FFN, ASL in place of binary cross-entropy, and the coherence rule. With 1 seed the HGCLR design scores about 2 points of macro AP above the base, on the validation data and on the real test set; the gap belongs to these 5 parts together.
- **The ensemble of 3 seeds adds about as much again**, group 4, Section 6.7. It is the final model.
- **The encoder size matters more than the input length**, group 3: the 610m on 1,024 tokens scores above the 210m on 2,048 tokens, on the validation data and on the real test set. The max rule in the loss hurts: the 610m trained with it loses about 2 points of macro AP against the 610m trained without a rule. The 2 610m runs differ in their soft values and article-to-topic attention as well, so part of the loss may belong to those settings.

Table 6.6: **Groups 3 and 4: pruned gold set, rebuilt pipeline**, full corpus. Compare rows within a group only. One seed unless stated.

				Validation		Real test		
Model	Encoder	Input	Rule	Macro AP	Micro AP	Macro AP	Micro AP	Violation rate
Group 3: all topics in training; real test set before trimming, 10,000 articles								
HGCLR design	210m	2,048	avg-max	0.810	0.892	0.654	0.779	0
Earlier HGCLR design	610m	1,024	none	**0.824**	**0.897**	**0.685**	**0.796**	0.009
HGCLR design	610m	1,024	max	0.799	0.885	0.646	0.774	0
Group 4: 6 weak topics folded; real test set after trimming, 9,701 articles								
Base: no graph encoder, positive view, token FFN, ASL or rule	210m	2,048	none	0.785	0.879	0.630	0.764	0.007
HGCLR design, 1 seed, mean of 3	210m	2,048	avg-max	0.804	0.890	0.652	0.780	0
HGCLR design, ensemble of 3 seeds: final model	210m	2,048	avg-max	**0.823**	**0.903**	**0.683**	**0.803**	0

6.6 Choice of the coherence rule

Table 6.7 gathers the evidence for the rules from both run indexes.

- **After training, the max rule lowers macro F1 and the avg-max rule raises it**, in most of the runs trained without a rule.
- **Inside the training loss, panel B, the max rule lowered macro AP in 2 pairs out of 3**: on the first gold set and on the full gold set with the 610m. On the gold subset it raised it. The avg-max rule has 1 pair, on the gold subset, where it did not lower macro AP. The 610m pair differs in 2 other settings as well, so it is weak evidence on its own.

The final model therefore trains and predicts with the avg-max rule. Chapter 8 lists what the record leaves open.

Table 6.7: The evidence for the rules. A: each rule applied after training to the saved scores of the 58 runs of the old platform trained without a rule, macro F1 on the validation data, **groups 1 and 2**. B: the rule inside the training loss, macro AP on the validation data; **each pair sits inside 1 group** of Table 6.1; a dash marks a pair that was not run. *The 2 runs differ in their soft values and article-to-topic attention as well.

Group	Runs	No rule	Max	Avg-max
A. Rule applied after training, macro F1				
1 and 2	58 runs, mean	0.494	0.492	0.496
	runs above no rule	–	4 of 58	52 of 58
B. Rule inside the training loss, macro AP				
2	frozen + head, BCE, ensemble	0.717	0.702	–
3, subset	HGCLR, 210m, 1 seed	0.738	0.744	0.743
3	HGCLR, 610m, 1 seed*	0.824	0.799	–

6.7 Seeds and the ensemble

Table 6.8 gives the 3 seeds of the final design, group 4.

- **The 3 seeds score almost the same**: the seed changes macro AP by less than 0.5 point.
- **Their ensemble scores above each seed**, by about 2 points of macro AP on the validation data and 3 on the real test set, at 3 times the inference cost.

Table 6.8: The final design with 3 seeds and their ensemble. **Group 4**, full corpus.

	Validation			Real test	
Seed	Macro AP	Micro AP	Macro F1	Macro AP	Micro AP
42	0.805	0.890	0.719	0.654	0.781
43	0.804	0.890	0.721	0.650	0.779
44	0.805	0.890	0.719	0.653	0.782
ensemble	**0.823**	**0.903**	**0.738**	**0.683**	**0.803**

Chapter 7

Error analysis

The metrics of this chapter skip 32 topics that the model learns, Table 7.1. They come from a review of the weakest topics one by one, on the dashboard and in the per-topic tables of the real test set. Each had a reason: the topic is unlearnable, since it overlaps with its neighbours and scores low on the generated data too; or it shifts, fine on generated articles and weak on real press; or the real test set holds too few of its articles to measure it. All 32 are leaves whose parent still catches their articles, so the team ruled that the model keeps learning them and the metrics skip them. The review used the real test scores, so the test scores over the 291 topics are optimistic; Table 7.2 gives the scores over all 323 topics as well.

Table 7.1: The 32 topics the model learns and the metrics skip, by the reason found for each.

Group	Topics	Meaning	Examples
Unlearnable	11	weak on the generated validation data as well: the topic overlaps with its neighbours	*record and achievement*, *civil rights*, *business service*
Shift	9	fine on the generated data, weak on real press	*birthday*, *surveillance*, *freedom of the press*
Thin	12	under 30 articles in the real test set	*coup d'etat*, *philosophy*, *volunteering*

7.1 The errors by level

Table 7.2 splits the scores of the final model by level, on the validation data and on the real test set. 2 patterns hold. **The deeper the level, the lower the score**: level 1 is almost solved, level 3 is where the model errs. And each level scores lower on real articles than on generated ones, with the largest drop at levels 2 and 3. On the validation data the model predicts more topics per article than the LLMs gave, so precision is the weak side there.

7.2 Errors on the validation data: the model or the LLM topics?

On the validation data a disagreement between the model and the LLM topics has 3 possible causes: the model is wrong, the LLM topics are wrong, since they are votes and not truth, or the article sits between 2 similar topics and either answer holds.

The weakest topics are the ones with the fewest training articles. Figure 7.1 shows the view of my dashboard per topic for the final model on the validation data: on the left, each topic's AP against its number of training articles; on the right, the topics ranked by AP inside each level. The topics below the macro AP line are level 2 and 3 topics with a few hundred

Table 7.2: The final model by level. Macro AP, and precision and recall over all article-topic pairs at the threshold of 0.5.

		Validation, generated			Test, real		
Level	Topics	Macro AP	Precision	Recall	Macro AP	Precision	Recall
1	17	0.965	0.859	0.943	0.907	0.821	0.803
2	106	0.848	0.731	0.889	0.711	0.689	0.742
3	168	0.792	0.615	0.821	0.642	0.599	0.667
all, scored	291	0.823	0.725	0.884	0.683	0.705	0.743
all, with the 32 skipped	323	0.805	0.721	0.879	0.641	0.701	0.737
Topics per article, LLM / model		12.8 / 15.6			11.3 / 11.9		

articles. The figure leaves out the 32 topics that the model learns and the metrics skip, as Table 7.2 does. So the errors sit where the data is thin, and the question is what kind of error they are.

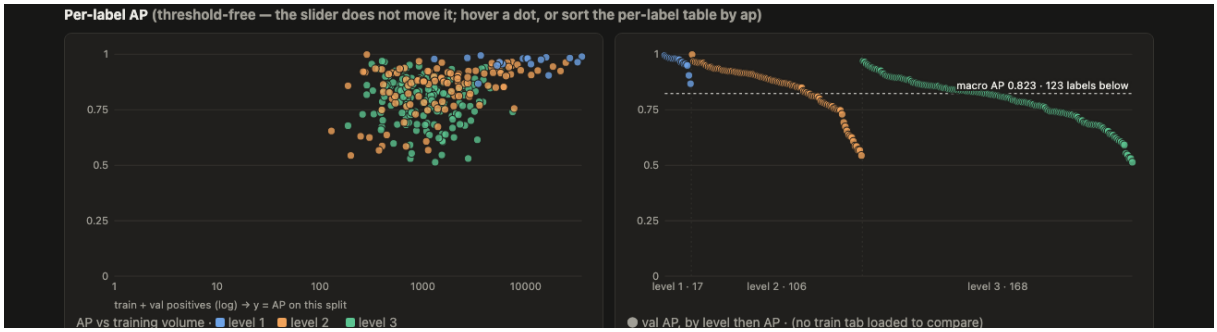


Figure 7.1: **The per-topic view of the dashboard**, final model, validation data, over the 291 scored topics. Left: AP of each topic against its number of training articles, log scale, 1 dot per topic, coloured by level. Right: the topics ranked by AP inside each level, with the macro AP line.

The goal here is to explore the disagreements and find where the errors are. To tell the causes apart I read the worst articles: the articles with the most wrong level 1 topics at the threshold of 0.5, and then the largest gap between the model's scores and the LLM topics. Table 7.3 gives 1 article per cause, and Table 7.4 the full topic list of 1 of them. Wrong LLM topics need a judge pass on levels 2 and 3; model errors need more articles on the rare topics; hard cases stay.

An LLM judge then scored a sample of the disagreements, Table 7.5. A topic the model adds is wrong in most cases. A topic the model misses is a wrong LLM topic in most cases. So the precision gap is real, and the recall gap is in large part noise in the topics. The true recall of the model is therefore higher than the measured one, if the judge is right, which is good news, since the product puts recall first, Section 1.1. The audit ran on an earlier run, with 1 LLM judge, the newest and strongest model available, and no human check.

7.3 Errors on the real test set: a different mix of topics

An LLM wrote the training and validation articles; the test articles are real French press that the team scraped online, and many scraped articles stop short, at a paywall or a preview. Table 7.6 gives the share of articles with each level 1 topic. **Politics and economy fill the generated articles; real press has far more lifestyle, weather, environment and arts.** The model scores lower on these topics, and it gives *human interest* to many more

Table 7.3: 4 causes of a disagreement on the validation data, 1 article each. The LLM topics and the model's topics at level 1 and 2, where they differ.

Cause	Article	LLM topics vs model	What the text says
The LLM topics are wrong	*Orange Metalic: 3 headliners for a sold-out night*	LLMs: weather, with phenomena, warning, forecast; model: lifestyle, arts, economy	a concert report of 2,500 people; the weather is 1 line on a local alert at the end of the night
The LLM topics miss a topic	*A retired woman from Ajaccio knits a 21 km scarf*, Table 7.4	LLMs: no *senior citizens*; model: 0.96	the text is about a 72-year-old retired woman
The model is wrong	*French stand-up comedians tour the country this autumn*	LLMs: arts and entertainment, theatre; model adds economy, business enterprise, start-up at 0.5 to 0.6	a tour of shows; the text names the organisers, and the business is not the subject
Hard for a human too	the same scarf article	LLMs: *fashion*, 2 votes; model: 0.05	a knitted scarf as a record; fashion fits the object, not the story

Table 7.4: A validation article where the model and the LLM topics disagree. *Une retraitée d'Ajaccio tricote une écharpe de 21 kilomètres, chantier textile inédit en Corse*: a 72-year-old retired woman finishes a scarf 21 km long; the body stops at a paywall after 2 sentences, as the generator cuts a share of the articles on purpose, the way scraped news arrives. Votes: LLMs out of 3 that gave the topic; 0 means no LLM did.

Topic	Votes	Score
human interest	3	1.00
human interest > record and achievement	3	0.66
lifestyle and leisure	3	0.49
lifestyle and leisure > leisure	2	0.31
arts, culture, entertainment and media > arts and entertainment > fashion	2	0.05
society > demographic group > senior citizens	0	0.94
society > communities	0	0.71

Table 7.5: Blind LLM judge on validation disagreements, earlier run of the same design: 150 random validation articles, 1,066 disputed topics. 1 judge, name of the judge LLM to fill in, no human check. The rest of each row is undecided.

Disagreement	Model wrong	LLM topics wrong	Undecided
model adds a topic the LLMs did not give	88 %	6 %	6 %
model misses a topic the LLMs gave	26 %	56 %	18 %

real articles than the LLMs do.

Table 7.6: Share of articles that carry a level 1 topic, in the generated validation data and in the real test set, and the model's results on the real test set at the threshold of 0.5. Topics with the largest change.

	LLM topics		Model, real test set		
Level 1 topic	Generated	Real	Predicted	Precision	Recall
politics and government	60 %	38 %	35 %	0.92	0.84
economy, business and finance	46 %	36 %	34 %	0.86	0.82
crime, law and justice	27 %	17 %	15 %	0.88	0.78
lifestyle and leisure	6 %	28 %	22 %	0.89	0.69
environment	9 %	22 %	19 %	0.92	0.79
weather	2 %	13 %	9 %	0.97	0.65
arts, culture, entertainment and media	16 %	24 %	25 %	0.78	0.82
human interest	29 %	33 %	52 %	0.59	0.92

3 steps follow for the next model. Work on precision, since most topics the model adds are wrong and many topics it misses are noise in the votes. Choose the checkpoint and the threshold on real articles, since the generated data has another topic mix. Filter index pages and digests out of the real input before the model sees them.

Chapter 8

Limits

4 limits of the work stand out.

- **No run tests the depth of the head.** The head is a 3-layer MLP from the first frozen baseline on. I did not compare it with the single linear layer of HGCLR or with other depths.
- **No direct test backs the choice of the avg-max rule.** I compared the max rule with no rule, and the max rule lost, Table 6.7. I did not compare the avg-max rule with no rule on the full gold set. So the record shows that the max rule hurts, not that the avg-max rule helps.
- **The avg-max loss has 1 open numeric case.** A logit below about -744 gives a score of 0 in float64, so its $\log p$ is minus infinity and the loss of the batch is not a number. A guard skips such a batch, which happened once in the final runs; a clamp on the logits would remove the case.
- **The easy data set hides the gain of HGCLR on rare topics.** It folded its rare topics into their parents, and HGCLR gains most on rare topics, in its paper and in my best run. So no run on the easy data set measures HGCLR where it should gain most.

Chapter 9

Side work: relevance rerankers for thematic search

9.1 Problem and data

9.2 zerank cross-encoders run locally

9.3 GLiNER2 as a zero-shot classifier

9.4 Findings

Chapter 10

Conclusion

The internship built a model that tags French press articles with the topics of the IPTC tree, trained on generated articles and tested on real ones. 3 parts of the work are new.

- **HGCLR rebuilt on EuroBERT, with 4 changes.** Attention pooling replaces the first token that EuroBERT lacks, an article-to-topic attention block brings the topic tree to the token states at inference, a deeper head scores the article, and the asymmetric loss replaces binary cross-entropy. An ensemble of 3 seeds is the final model.
- **The avg-max rule.** It keeps the predicted scores coherent along the tree, as the max rule of C-HMCNN does, and moves a parent and its child toward each other instead of copying the child.
- **A run platform and an error dashboard.** Each run keeps its commit, its configuration and its metrics in one index, so the choices of the project compare on the same footing, and the dashboard traces an error to a topic, an article or a label.

I learned 3 things from this project that no course taught me.

- **How to work with an AI agent, and how to lead it.** The agent ran the project with me: it planned the work, drafted and refactored the code, searched the research, wrote the tests and kept the run index, and I used it to study. The skill is on my side of the dialogue: a clear brief, a fixed style, a review of each output, and a record of each decision. With this guidance the code and the report stay readable for a human; without it, the agent writes long text in a polished style that hides what the project did.
- **A better data set matters more than a better model.** The gap between generated and real articles, the noise in the LLM topics and the rare topics set the scores more than any block of the model. The gold set with its votes, and the soft value for a topic with 1 vote, gave larger gains than most changes of the architecture.
- **How to read new research, rebuild it, and pass it on.** HGCLR made sense on paper and fell apart on the first run; rebuilding the paper's model as published, before changing anything, was what made the later changes measurable, and it sent me back to the core of machine learning, losses, gradients and schedules. Sharing what I learned, in reviews with my mentors and the other interns, in a workshop on attention for the team, and in this report, was the part that fixed the ideas in place.

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Appendix A

Method details

A.1 The Graphormer label-attention baseline

The third model is the first one that sees the tree. Figure A.1 shows its 2 sides. The topic side starts from the tree. Each topic gets a vector from the frozen EuroBERT embedding of its name and definition, plus a learned vector for its number of neighbours. A Graphormer then runs attention between the topics, and the attention score of 2 topics carries a learned bias set by their distance in the tree. The article side runs the article through EuroBERT. The 2 sides meet in the label attention of HiAGM-LA [15]: each topic attends over the tokens of the article and gathers its own view of it, and a scorer per topic turns that view into the topic's score. The encoder can stay frozen or train with the rest.

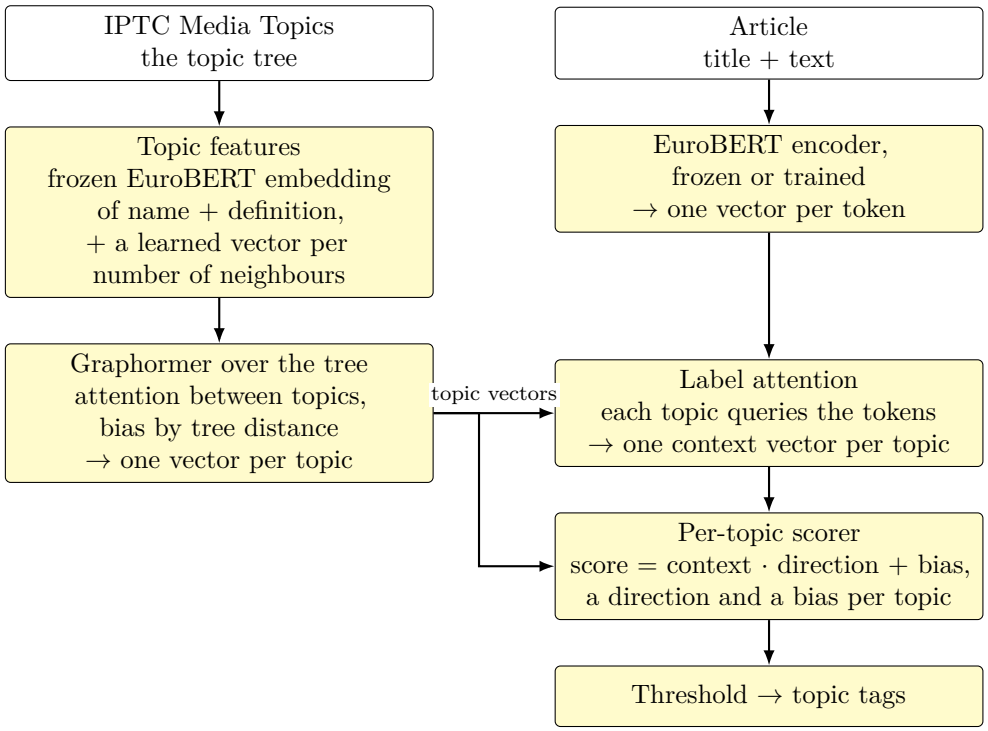


Figure A.1: The Graphormer label-attention baseline. The topic side on the left and the article on the right meet in the label attention, where each topic reads the article for itself. White: inputs. Yellow: the blocks of this baseline.

A first version scored all topics with one shared scorer and gave each topic the same score; **the scorer per topic fixes this**.

A.2 Cheaper ways to train the encoder: top layers and LoRA

2 cheaper options sit between the frozen encoder and the full fine-tune. The first trains the top layers of the encoder and freezes the lower ones. The frozen layers keep their pretrained features, and AdamW holds no state for them, so the run is lighter. Early in the exploration I trained 2 or 3 top layers, without a record, and saw little change from the frozen model. Later, with HGCLR, training the top 6 layers cut the training time and scored above the frozen model. I have no clean comparison with the full fine-tune: the 2 runs used different settings for the positive samples, so the gap between them has 2 causes. I kept the option for cheap test runs; the final model trains the whole encoder.

The second option is LoRA [5]. The pretrained weights stay fixed. Each attention projection gets a pair of small low-rank matrices, and the gradient reaches these matrices and the head, under 1 % of the parameters. I tried it in the early exploration, without a record: LoRA brought no clear gain over the frozen model, and I dropped it.

A.3 The avg-max rule: gradient and implementation

The avg-max rule cannot run on the logits: the mean of 2 scores is not the sigmoid of the mean of their logits. So it runs on the scores, as the loop of Section 4.4.1, per article and per tree. Each step of the loop is a mean, which is linear, and the choice of which pairs to average carries no gradient: a comparison of 2 scores is a yes-or-no decision, and its derivative is 0, so autograd on the tensor version gives the same gradient as the tape, through the means alone. The forward pass therefore records the pairs it averaged, as a tape, and the backward pass replays the same means in reverse order on the gradient. The implementation went through 3 versions. The first ran the loop on tensors, with a `torch.where` and 2 index writes per step, so autograd tracked each step, fired or not, and the backward pass walked them all; it was far too slow. The second ran the loop in plain code and recorded the means as a matrix M per article, then computed $P = Mw$ under autograd, one linear operation per batch. The third, in the final model, is the tape: it records the pairs alone and replays them, with no matrix. The loop runs in float64 on the CPU, compiled with numba, and takes about as long as the max rule on the GPU.

The loss needs $\log p$ and $\log(1-p)$. **The loop therefore takes 2 inputs, the score of a topic being present, $\sigma(s)$, and the score of it being absent, $\sigma(-s)$, and applies each mean to both**, so that $\log(1-p)$ needs no subtraction in floating point.

A.4 Training settings

The final model trains the whole encoder with the settings of Table A.1. The optimizer is AdamW with 3 learning rates: a low one for the encoder, which has to keep its language skill, a middle one for the graph encoder, and a high one for the blocks that start from scratch, the attention, the pool, the head and the projection. The learning rate warms up over the first steps and then follows a cosine down to a floor. The batch is 16 articles, whatever the GPU. A large GPU holds all 16 in 1 step. A small one holds the same batch through gradient accumulation: it runs the 16 articles in several smaller steps, sums their gradients, and updates the weights once. A small script picks the split from the GPU's memory and an estimate of the memory one step needs. A topic with 1 vote out of 3 enters the loss as a soft value, Section 3.3. Such a topic has $y = 0.5$ in the loss instead of $y = 1$, since 1 vote out of 3 is weak evidence, Table A.1. On the validation and test data each topic is present or absent, $y = 1$ or $y = 0$. On the gold subset this soft value and the input format with the article-to-topic attention of Section 4.3.3 were the 2 changes that raised the score most. A run stops when the validation loss has not improved for a few evaluations, and the checkpoint with the best macro average precision is the one kept.

Table A.1: Training settings of the final model.

Setting	Value	Note
Trained parameters	all layers of the encoder, all blocks	
Optimizer	AdamW, weight decay 0.01	
Learning rate, encoder	3×10^{-5}	
Learning rate, graph encoder	1×10^{-4}	
Learning rate, other blocks	1×10^{-3}	attention, pool, head, projection
Schedule	warm-up 6 %, then cosine to 10 % of the peak	
Gradient clipping	1.0	
Effective batch	16 articles	micro-batch × accumulation, from the GPU's memory
Precision	bf16 autocast	fp32 master weights
Input length	2,048 tokens	the configuration file allows 4,096
Soft value	0.5 for a topic with 1 vote	2 votes or more count as 1
Early stop	validation loss, patience 3 evaluations	2 on the full corpus, 2 evaluations per epoch
Checkpoint kept	best macro average precision	

Appendix B

Metric definitions

The model gives one score per topic, and I read the scores in 2 ways. With a threshold, each score becomes a yes or a no, and the usual counts follow: **recall, the share of the article's topics the model found, and precision, the share of the model's topics that are right**. F1 is their harmonic mean. With TP the topics found, FP the topics predicted but wrong and FN the topics missed,

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}. \tag{B.1}$$

Without a threshold, **average precision (AP) ranks the scores and rewards a model that puts the right topics first**, whatever the threshold. It walks down the ranked list and sums the precision at each point where a right topic appears, weighted by the recall gained there:

$$AP = \sum_k \left(R_k - R_{k-1}\right) P_k, \tag{B.2}$$

where P_k and R_k are the precision and the recall of the first k scores.

Each of these comes in 2 flavours. Micro pools all decisions of all articles into one count, so the frequent topics dominate. Macro computes the score per topic and takes the mean, so a rare topic counts as much as a frequent one. For precision, with L topics and TP_j, FP_j the counts of topic j,

$$P_{\text{micro}} = \frac{\sum_{j=1}^{L} TP_j}{\sum_{j=1}^{L} (TP_j + FP_j)}, \qquad P_{\text{macro}} = \frac{1}{L} \sum_{j=1}^{L} P_j, \tag{B.3}$$

and the same for recall, F1 and AP.

The violation rate V is the share of predicted child topics whose parent is not predicted. With $\hat{y}_{ij} \in \{0, 1\}$ the prediction for article i and topic j, and C the set of topics that have a parent,

$$V = \frac{\sum_i \sum_{j \in C} \hat{y}_{ij} \left(1 - \hat{y}_{i,\text{parent}(j)}\right)}{\sum_i \sum_{j \in C} \hat{y}_{ij}}. \tag{B.4}$$

A coherent model has $V = 0$. The run index records 2 more numbers that the tables of this report do not show. **Recall at k is the share of an article's topics y_i among its k highest scores**, averaged over the N articles:

$$R@k = \frac{1}{N} \sum_{i=1}^{N} \frac{|\, y_i \cap \text{top}_k(i) \,|}{|\, y_i \,|}. \tag{B.5}$$

The break value B measures the breaks of the tree on the raw scores. For each parent p, take each child c that scores above it and square the gap $p_c - p_p$. Sort the gaps under one

parent from large to small, and give the r-th the weight $1/r$. B is the sum over the parents, averaged over the articles:

$$B = \frac{1}{N} \sum_{i=1}^{N} \sum_{p} \sum_{r} \frac{1}{r}\, g_r^2, \qquad g_r \text{ the } r\text{-th largest gap under } p \text{ in article } i. \tag{B.6}$$

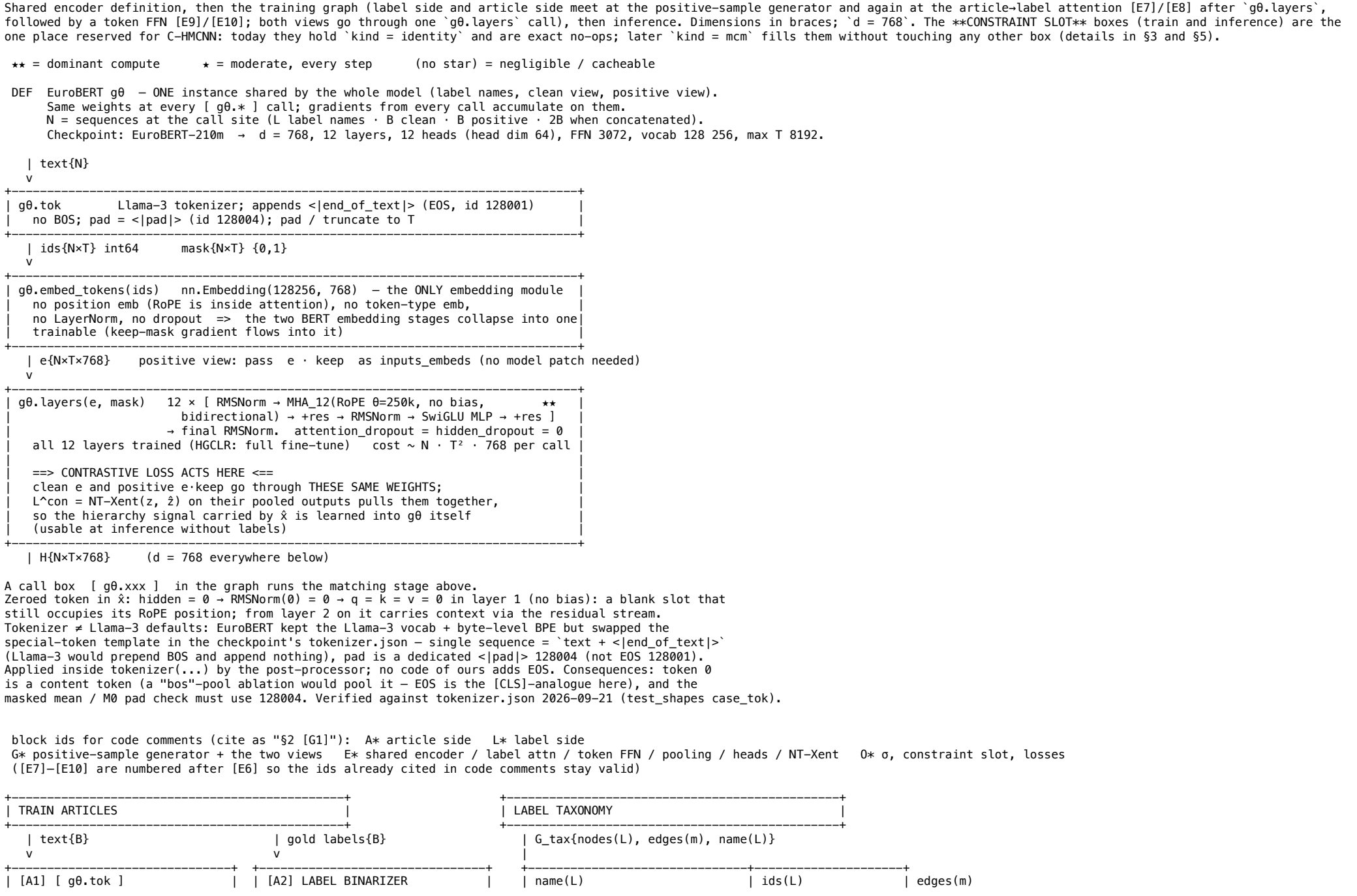
The square makes one real break count more than many small ones, and the weights $1/r$ stop one low parent from counting once per child. $B = 0$ means coherent scores. The run index of the old platform records it as `broken`.

Appendix C

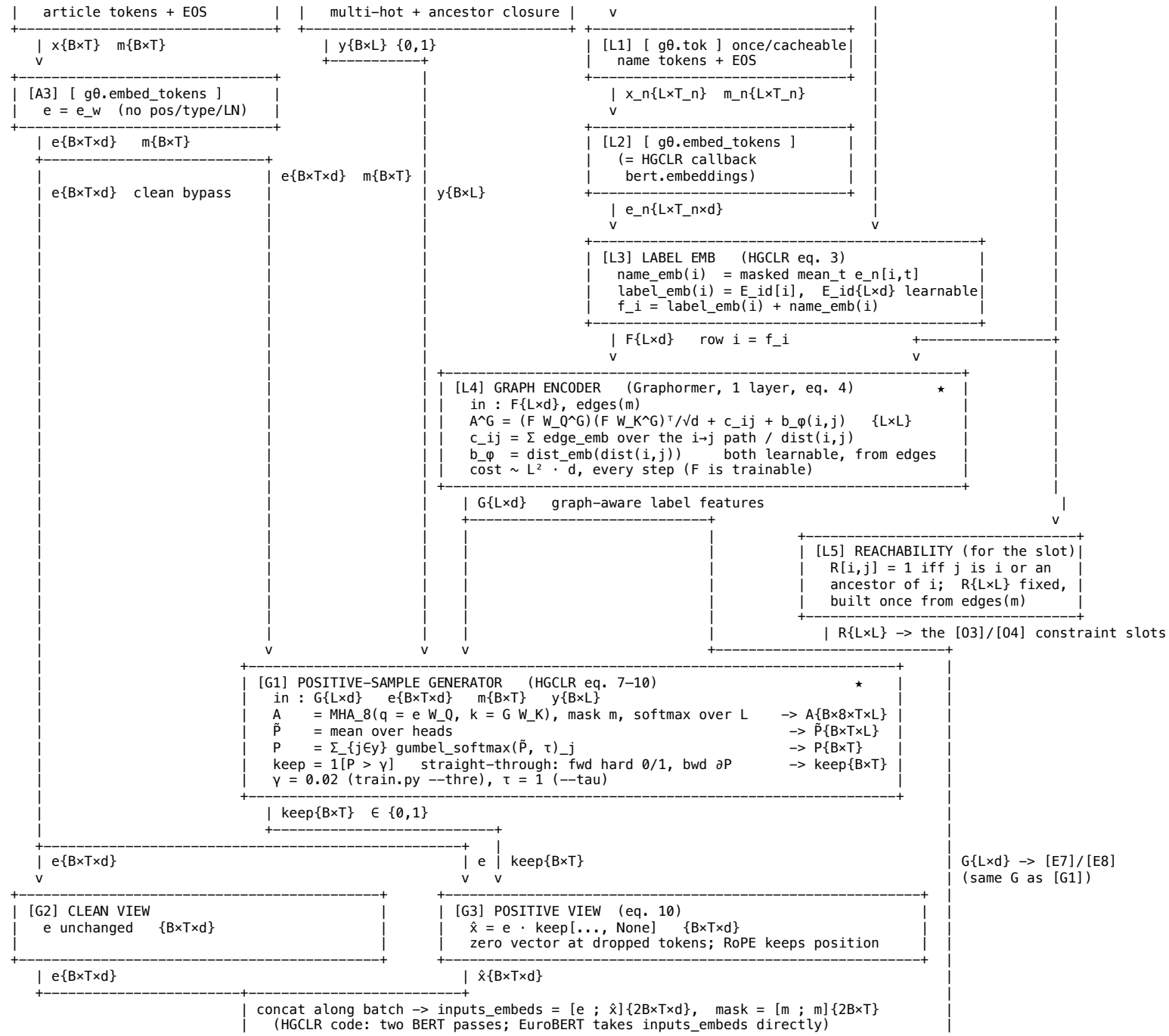
Full architecture graph

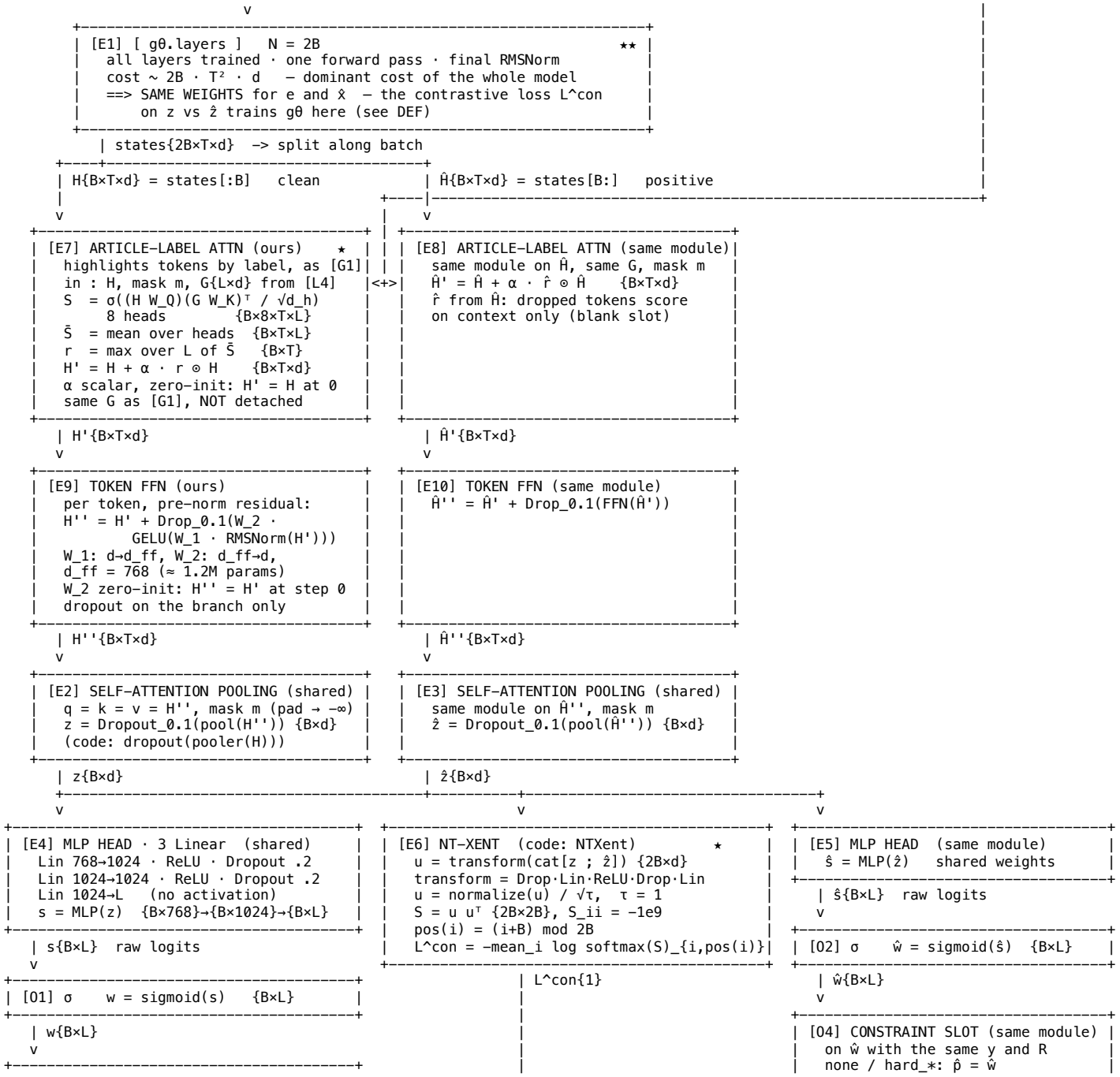
The pages that follow copy the architecture graph from the documentation of the code: the encoder, the training graph and the inference graph, block by block.

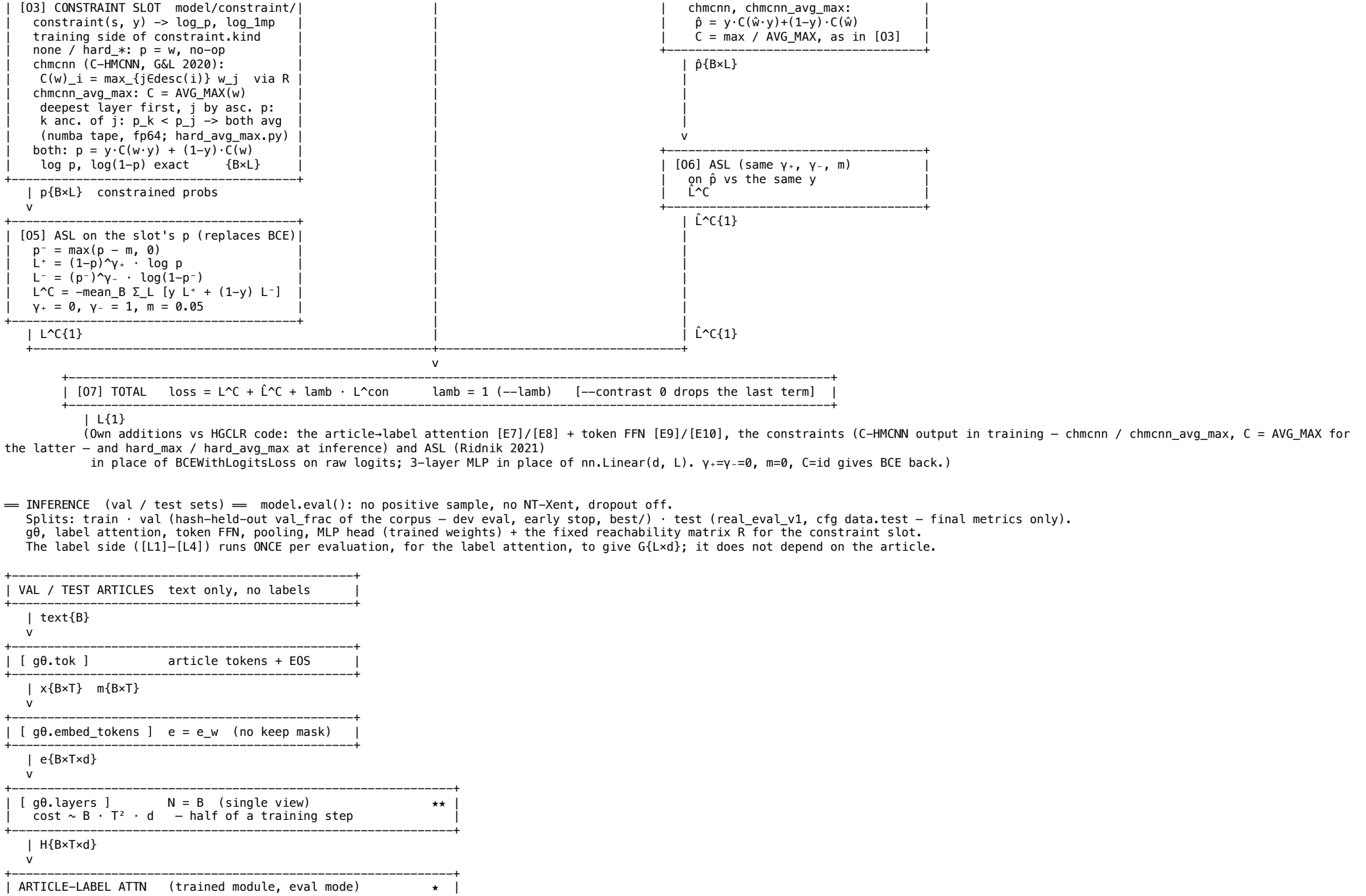
Shared encoder definition, then the training graph (label side and article side meet at the positive-sample generator and again at the article→label attention [E7]/[E8] after `g0.layers`, followed by a token FFN [E9]/[E10]; both views go through one `g0.layers` call), then inference. Dimensions in braces; `d = 768`. The **CONSTRAINT SLOT** boxes (train and inference) are the one place reserved for C-HMCNN: today they hold `kind = identity` and are exact no-ops; later `kind = mcm` fills them without touching any other box (details in §3 and §5).



A call box [g0.xxx] in the graph runs the matching stage above.
Zeroed token in x̂: hidden = 0 → RMSNorm(0) = 0 → q = k = v = 0 in layer 1 (no bias): a blank slot that still occupies its RoPE position; from layer 2 on it carries context via the residual stream.
Tokenizer ≠ Llama-3 defaults: EuroBERT kept the Llama-3 vocab + byte-level BPE but swapped the special-token template in the checkpoint's tokenizer.json — single sequence = `text + <|end_of_text|>` (Llama-3 would prepend BOS and append nothing), pad is a dedicated <|pad|> 128004 (not EOS 128001). Applied inside tokenizer(...) by the post-processor; no code of ours adds EOS. Consequences: token 0 is a content token (a "bos"-pool ablation would pool it — EOS is the [CLS]-analogue here), and the masked mean / M0 pad check must use 128004. Verified against tokenizer.json 2026-09-21 (test_shapes case_tok).

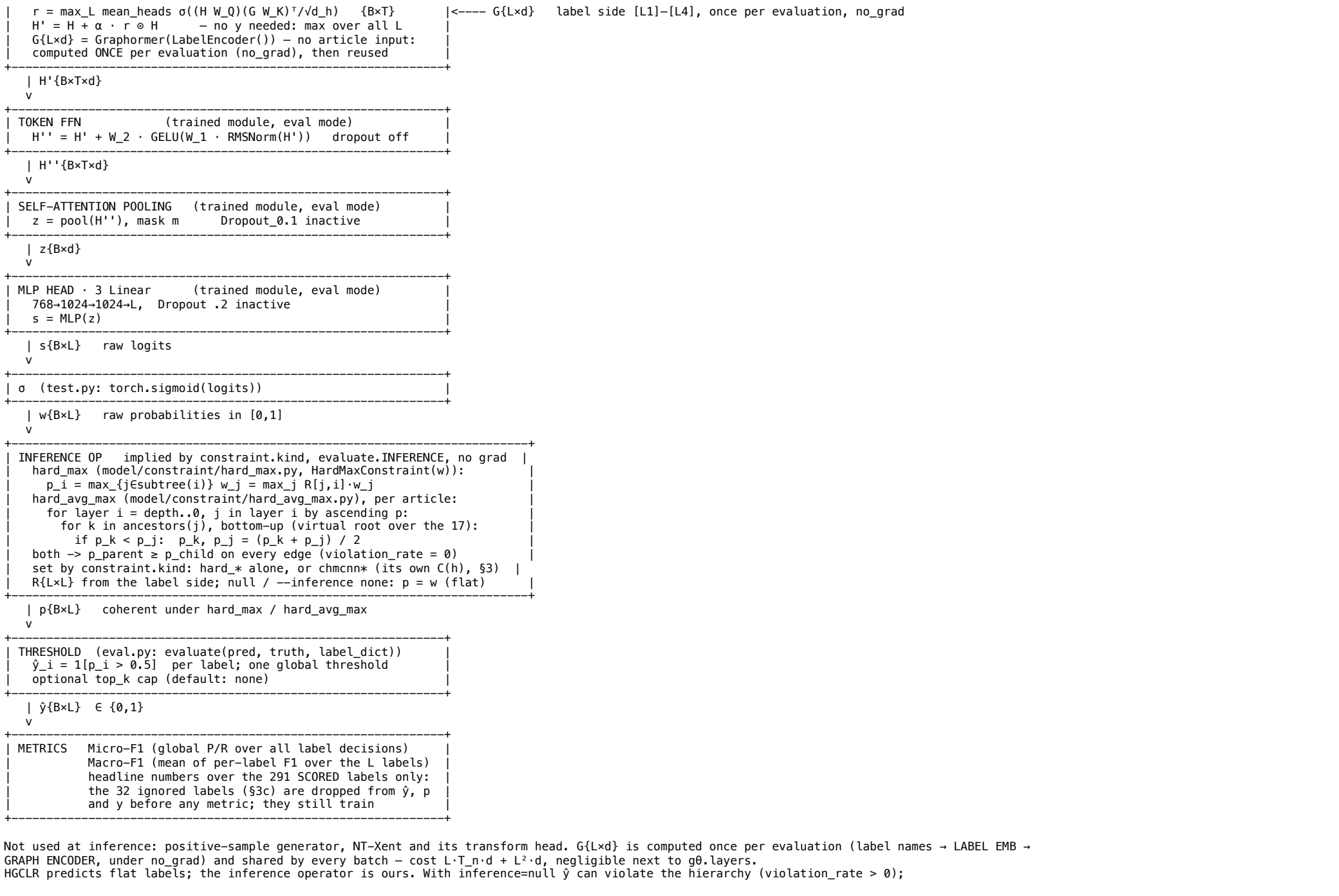

article tokens + EOS
multi-hot + ancestor closure
x{B×T} m{B×T}
y{B×L} {0,1}
[A3] [gθ.embed_tokens]
e = e_w (no pos/type/LN)
e{B×T×d} m{B×T}
e{B×T×d} clean bypass
[L1] [gθ.tok] once/cacheable
name tokens + EOS
x_n{L×T_n} m_n{L×T_n}
[L2] [gθ.embed_tokens]
(= HGCLR callback
bert.embeddings)
e_n{L×T_n×d}
[L3] LABEL EMB (HGCLR eq. 3)
name_emb(i) = masked mean_t e_n[i,t]
label_emb(i) = E_id[i], E_id{L×d} learnable
f_i = label_emb(i) + name_emb(i)
F{L×d} row i = f_i
[L4] GRAPH ENCODER (Graphormer, 1 layer, eq. 4)
in : F{L×d}, edges(m)
A^G = (F W_Q^G)(F W_K^G)ᵀ/√d + c_ij + b_φ(i,j) {L×L}
c_ij = Σ edge_emb over the i→j path / dist(i,j)
b_φ = dist_emb(dist(i,j)) both learnable, from edges
cost ~ L² · d, every step (F is trainable)
G{L×d} graph-aware label features
[L5] REACHABILITY (for the slot)
R[i,j] = 1 iff j is i or an
ancestor of i; R{L×L} fixed,
built once from edges(m)
R{L×L} -> the [O3]/[O4] constraint slots
[G1] POSITIVE-SAMPLE GENERATOR (HGCLR eq. 7–10)
in : G{L×d} e{B×T×d} m{B×T} y{B×L}
A = MHA_8(q = e W_Q, k = G W_K), mask m, softmax over L -> A{B×8×T×L}
P̃ = mean over heads -> P̃{B×T×L}
P = Σ_{j∈y} gumbel_softmax(P̃, τ)_j -> P{B×T}
keep = 1[P > γ] straight-through: fwd hard 0/1, bwd ∂P -> keep{B×T}
γ = 0.02 (train.py --thre), τ = 1 (--tau)
keep{B×T} ∈ {0,1}
e{B×T×d}
e | keep{B×T}
G{L×d} -> [E7]/[E8]
(same G as [G1])
[G2] CLEAN VIEW
e unchanged {B×T×d}
[G3] POSITIVE VIEW (eq. 10)
x̂ = e · keep[..., None] {B×T×d}
zero vector at dropped tokens; RoPE keeps position
e{B×T×d}
x̂{B×T×d}
concat along batch -> inputs_embeds = [e ; x̂]{2B×T×d}, mask = [m ; m]{2B×T}
(HGCLR code: two BERT passes; EuroBERT takes inputs_embeds directly)


[E1] [gθ.layers] N = 2B **
all layers trained · one forward pass · final RMSNorm
cost ~ 2B · T² · d — dominant cost of the whole model
==> SAME WEIGHTS for e and x̂ — the contrastive loss L^con
on z vs ẑ trains gθ here (see DEF)
states{2B×T×d} -> split along batch
H{B×T×d} = states[:B] clean
Ĥ{B×T×d} = states[B:] positive
[E7] ARTICLE-LABEL ATTN (ours) ★
highlights tokens by label, as [G1]
in : H, mask m, G{L×d} from [L4]
S = σ((H W_Q)(G W_K)ᵀ / √d_h)
8 heads {B×8×T×L}
S̄ = mean over heads {B×T×L}
r = max over L of S̄ {B×T}
H' = H + α · r ⊙ H {B×T×d}
α scalar, zero-init: H' = H at 0
same G as [G1], NOT detached
[E8] ARTICLE-LABEL ATTN (same module)
same module on Ĥ, same G, mask m
Ĥ' = Ĥ + α · r̂ ⊙ Ĥ {B×T×d}
r̂ from Ĥ: dropped tokens score
on context only (blank slot)
H'{B×T×d}
Ĥ'{B×T×d}
[E9] TOKEN FFN (ours)
per token, pre-norm residual:
H'' = H' + Drop_0.1(W_2 ·
GELU(W_1 · RMSNorm(H')))
W_1: d→d_ff, W_2: d_ff→d,
d_ff = 768 (≈ 1.2M params)
W_2 zero-init: H'' = H' at step 0
dropout on the branch only
[E10] TOKEN FFN (same module)
Ĥ'' = Ĥ' + Drop_0.1(FFN(Ĥ'))
H''{B×T×d}
Ĥ''{B×T×d}
[E2] SELF-ATTENTION POOLING (shared)
q = k = v = H'', mask m (pad → −∞)
z = Dropout_0.1(pool(H'')) {B×d}
(code: dropout(pooler(H)))
[E3] SELF-ATTENTION POOLING (shared)
same module on Ĥ'', mask m
ẑ = Dropout_0.1(pool(Ĥ'')) {B×d}
z{B×d}
ẑ{B×d}
[E4] MLP HEAD · 3 Linear (shared)
Lin 768→1024 · ReLU · Dropout .2
Lin 1024→1024 · ReLU · Dropout .2
Lin 1024→L (no activation)
s = MLP(z) {B×768}→{B×1024}→{B×L}
[E6] NT-XENT (code: NTXent) ★
u = transform(cat[z ; ẑ]) {2B×d}
transform = Drop·Lin·ReLU·Drop·Lin
u = normalize(u) / √τ, τ = 1
S = u uᵀ {2B×2B}, S_ii = -1e9
pos(i) = (i+B) mod 2B
L^con = −mean_i log softmax(S)_{i,pos(i)}
[E5] MLP HEAD (same module)
ŝ = MLP(ẑ) shared weights
s{B×L} raw logits
ŝ{B×L} raw logits
[O1] σ w = sigmoid(s) {B×L}
[O2] σ ŵ = sigmoid(ŝ) {B×L}
w{B×L}
ŵ{B×L}
L^con{1}
[O4] CONSTRAINT SLOT (same module)
on ŵ with the same y and R
none / hard_*: p̂ = ŵ



(Own additions vs HGCLR code: the article→label attention [E7]/[E8] + token FFN [E9]/[E10], the constraints (C-HMCNN output in training — chmcnn / chmcnn_avg_max, C = AVG_MAX for the latter — and hard_max / hard_avg_max at inference) and ASL (Ridnik 2021)
in place of BCEWithLogitsLoss on raw logits; 3-layer MLP in place of nn.Linear(d, L). γ₊=γ₋=0, m=0, C=id gives BCE back.)

== INFERENCE (val / test sets) == model.eval(): no positive sample, no NT-Xent, dropout off.
Splits: train · val (hash-held-out val_frac of the corpus — dev eval, early stop, best/) · test (real_eval_v1, cfg data.test — final metrics only).
g0, label attention, token FFN, pooling, MLP head (trained weights) + the fixed reachability matrix R for the constraint slot.
The label side ([L1]–[L4]) runs ONCE per evaluation, for the label attention, to give G{L×d}; it does not depend on the article.



Not used at inference: positive-sample generator, NT-Xent and its transform head. G{L×d} is computed once per evaluation (label names → LABEL EMB → GRAPH ENCODER, under no_grad) and shared by every batch — cost L·T_n·d + L²·d, negligible next to gθ.layers.
HGCLR predicts flat labels; the inference operator is ours. With inference=null ŷ can violate the hierarchy (violation_rate > 0);

with hard_max ŷ is coherent (violation_rate = 0) — C-HMCNN's hard constraint C(w), so a C-HMCNN training side later changes training only.
constraint.kind (model/constraint/__init__.py): none (flat) · hard_max · hard_avg_max (post-processing only) · chmcnn (trained on MCM, scored with MCM = hard_max) · chmcnn_avg_max (trained on AVG_MAX, scored with hard_avg_max).